Contents

Chapter 1

Hydrodynamic interactions between two colloids near a wall

1.1 Introduction

So far in this discussion, I have only described the behaviour of a single colloid away from any confining boundary. While this system already exhibits rich physical phenomena, I now wish to extend the discussion to more realistic situations. From here, two evident paths of extension arise: the case a *suspension* of colloids, interacting with each other through indirect Hydrodynamic Interactions (HIs), and the case of a single colloid diffusing over a rigid, infinite boundary.

Even at long range, HIs play a strong role on the behaviour of colloids diffusing in a fluid [1] [2]. While far-field HIs are accurately described through relatively simple approximations, the near-field regime is dominated by lubrication effects and requires dedicated asymptotic descriptions. Naturally, the near-field regime has already been inverstigated in depth, and the lubrication theory is well established for the case of a single colloid over a plane [3], and for the case of two colloids in a bulk fluid [4][5]. However, the coupling between these two cases, that is to say the case of two colloids interacting with each other and with a confining boundary, is much less explored. A few experimental studies have explored the colloid-colloid-boundary coupling, but a formalisation of the problem is still lacking. In this Chapter, I investigate this coupling by following the experimental study of Dufresne et al. [6] and by extending the analysis through numerical simulations in the strongly confined regime, where the colloids are simultaneously very close to one another and to the boundary.

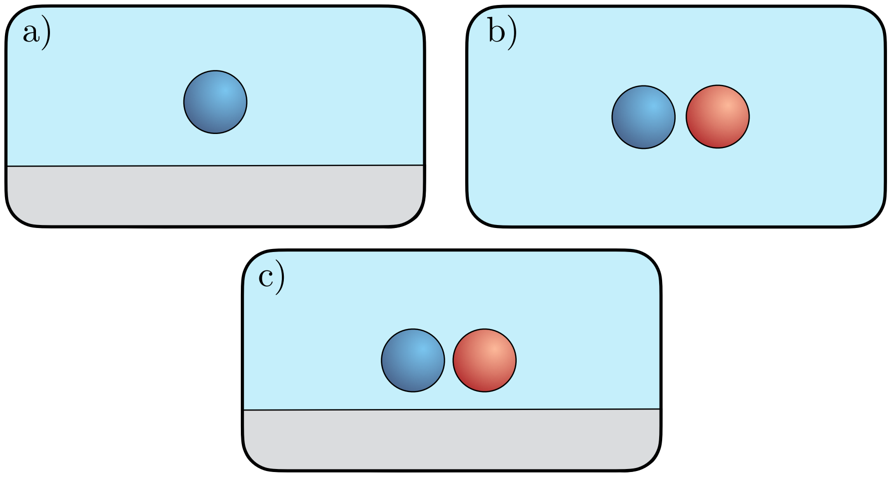

Figure 1.1: Schematic representation of the three cases of interest for this Chapter. a) A single colloid diffusing in a fluid over an infinite, hard boundary. b) Two colloids interacting with each other through hydrodynamic interactions in an unbounded fluid. c) A combination of the two first cases, where two colloids interact with each other and with an infinite, hard boundary.

Following the logical structure of Fig. 1.1, I open this Chapter with a full description of the diffusion of a colloid over a hard, infinite boundary, and show that there exist a few pitfalls in the formalisation of such a problem. Particular attention is paid to the stochastic description of Brownian motion near a wall, where the position dependence of the mobility tensor introduces issues in the formulation of the Langevin equation. I then move on to the case of two colloids in a bulk fluid and the various methods to simulate their interactions. Following these two cases, I present an algorithm build to simulate Brownian dynamics in confinement. This method is then applied to the investigation of the coupling between the wall-colloid and the colloid-colloid effects. Finally, I discuss how one may go beyond this study, in the deep-lubricated regime, and show how my numerical results compare to the previous study.

1.2 Colloid-plane lubrication theory

For the sake of completeness, I describe here all interactions that occur in an experimental setup of a silica colloid in a fluid near a silica plate. We take the plate normal to the vertical z axis, laying at $z = 0$. Over that plane is a fluid of density ρ, viscosity η, and thermal energy $k_B\Theta$ with k_B the Boltzmann constant. All parameters for the upcoming derivations are summarised in Table 1.1.

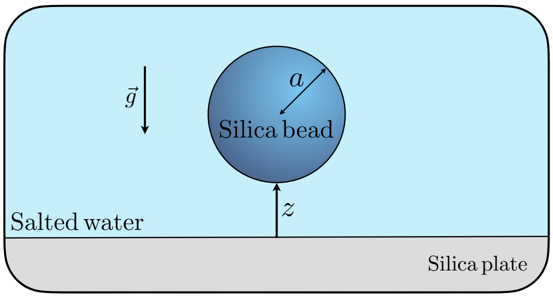

Figure 1.2: Schematic representation of a colloid of radius a at a distance z from an infinite plane.

In the fluid, we now submerge a colloidal silica particle of radius a, of the micron scale. Due to the no-slip boundary conditions at contact with the plane and the colloid's surfaces, we must modify the effective viscosity of the fluid as the colloid evolves in the z direction. This correction brings not only inhomogeneity along the z-axis but also anisotropy, as the x and y components of the friction coefficient differ from the z component.

Table 1.1: Typical parameters for a silica colloid suspended in water near a silica wall at room temperature. Values are indicative.

Symbol	Quantity	Typical value
a	Colloid radius	$0.5 - 2$ µm
z	Distance from wall centre-to-wall $a - 20a$	µm
ρ	Water density	10^3 kg.m^{-3}
η	Water viscosity	10^{-3} Pa.s
Θ	Temperature	298 K
k_B	Boltzmann constant	1.38×10^{-23} J.K^{-1}
$k_\mathrm{B}\Theta$	Thermal energy	4.1×10^{-21} J
$\eta_\perp(z)$	Effective perpendicular viscosity	η to $\gg \eta$
$\eta_\parallel(z)$	Effective parallel viscosity	η to $\gg \eta$
l_B	Boltzmann length	$0.1 - 10$ µm
l_D	Debye length	$1 - 100$ nm

In his seminal work[3], Brenner derived the exact expression for the effective viscosity felt by a sphere moving perpendicularly to a plane wall, $\eta_\perp(z)$, as:

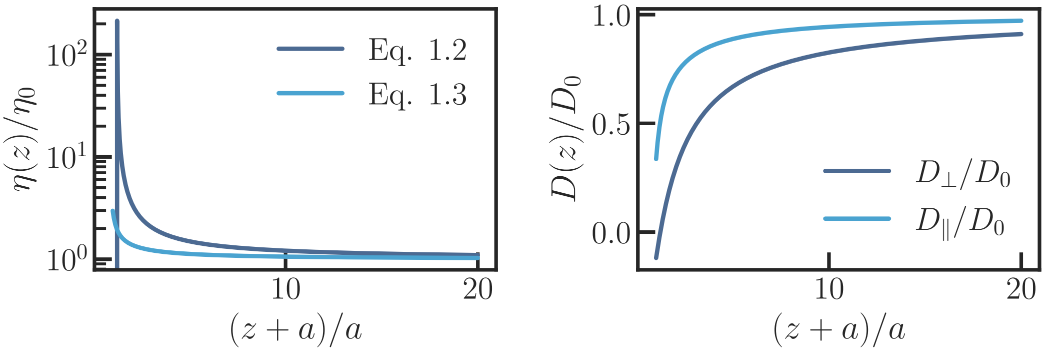

Figure 1.3: Effective viscosities and corresponding effective diffusion coefficients for a colloid diffusing over a plane. Left panel: Effective viscosities $\eta_\perp$ (navy) and $\eta_\parallel$ (sky blue) renormalised by the bulk viscosity η_0 as a function of the distance to the wall z. Right panel: Effective diffusion coefficients $D_\perp$ (navy) and $D_\parallel$ (sky blue) renormalised by the bulk diffusion coefficient D_0 as a function of the renormalised colloid center-to-wall distance $(z+a)/a$.

$$\eta_\perp(z) = \frac{4}{3}\eta\sinh(\alpha)\sum_{n=1}^{\infty}\frac{n(n+1)}{(2n-1)(2n+3)}\left\{\frac{2\sinh[(2n+1)\alpha]+(2n+1)\sinh(2\alpha)}{4\sinh^2[(n+\frac{1}{2})\alpha]-(2n+1)^2\sinh^2(\alpha)}-1\right\}, \tag{1.1}$$

with $\alpha = \cosh^{-1}\left(\frac{z+a}{a}\right)$. Because Brenner obtained the effective viscosity from solving the full Stokes equations with a no-slip boundary condition at the wall, it is worth noting that this expression is exact and valid for all values of $z/a > 0$, that is to say up until colloid/plane contact. However, for the sake of efficient numerical computations later on, we avoid infinite-series computations and prefer the use of Padé approximants [7], which were used by Honig [8], Bevan and Prieve [9] to approximate Brenner's result as:

$$\eta_\perp(z) \simeq \eta\frac{6z^2+2az}{6z^2-9az+2a^2}. \tag{1.2}$$

The parallel effective viscosity, on the other hand, was derived by Faxén [10] for $z/a > 0.1$ and is given by the following expression:

$$\eta_\parallel(z) \simeq \frac{\eta}{1-\frac{9}{16}\frac{a}{z}+\frac{1}{8}\left(\frac{a}{z}\right)^3-\frac{45}{256}\left(\frac{a}{z}\right)^4-\frac{1}{16}\left(\frac{a}{z}\right)^5}\,. \tag{1.3}$$

Both Eq. 1.2 and Eq. 1.3 demonstrate that the viscosity effectively increases as the colloid approaches the wall (Left panel of Fig. 1.3). In turn, the effective diffusion coefficients decrease (Right panel of Fig. 1.3).

Furthermore, these expressions can be used to simulate the Brownian motion of a colloid over a plane up until contact. We note that in the limit of $z/a \to 1$, $\eta_\perp$ captures the divergence close to the wall. This is an expected behaviour; as the colloid approaches the wall, the fluid in between is squeezed out and the lubrication forces diverge. However, $\eta_\parallel$ does not capture the weak logarithmic divergence expected as the surface-to-surface gap tends to zero. Other means of accurately capturing the divergence exist, which I mention down below.

1.2.1 Summary of potentials in the system

Furthermore, in order to write the correct overdamped Langevin equation for a simulation, we recall that the colloid is also subject to a number of forces, which we now carefully enumerate as they will be present throughout the remainder parts of this chapter.

Gravitational potential and buoyant mass

$$
\begin{aligned}
U_{\mathrm{grav}} &= \Delta m g z \ , && (1.4)\\
&= \frac{4}{3}\pi a^3(\rho_{\mathrm{colloid}} - \rho_{\mathrm{fluid}}) g z \ , && (1.5)
\end{aligned}
$$

where Δm is the buoyant mass of the colloid, g is the acceleration of gravity, and ρ_{colloid} and ρ_{fluid} are the mass densities of the colloid and the fluid, respectively. We may rewrite the above Eq. 1.5 as:

$$
U_{\mathrm{grav}} = \frac{k_{\mathrm{B}}\Theta}{l_{\mathrm{B}}} z \ , \tag{1.6}
$$

where l_{B} is Boltzmann's length, defined as $l_{\mathrm{B}} = \frac{k_{\mathrm{B}}\Theta}{\Delta m g}$. Boltzmann's length quantifies the distance over which gravitational potential energy is comparable to thermal energy.

Electrostatic potential and double-layer interactions

Because of the amphoteric nature of silica, both the infinite place and the colloid surfaces are able to ionise when in contact with water (Fig. 1.4). The surfaces themselves become negatively charged, which in turn attract adsorb ions in contact with the surface, in a dense and tightly bound Stern layer. Around this plane, a mobile, ionic "atmosphere"

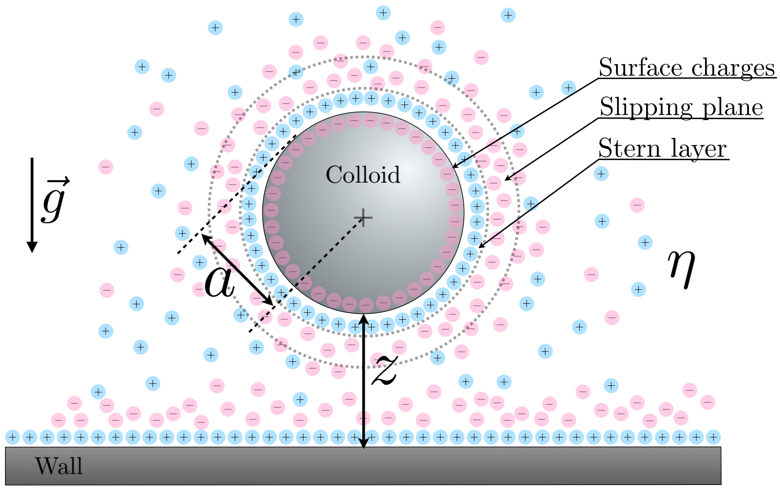

Figure 1.4: Schematic representation of a colloid of radius a at a distance z from an infinite plane. Both surfaces are charged, and thus, a double layer of counter-ions forms around each of them.

screens the electric potential of the colloid over distance with a characteristic length l_{D}, for which we provide the derivation below. This diffuse layer, existing for both surfaces at play in the system, is responsible for the electrostatic repulsion between the two bodies (plane and colloid). It is also worth mentioning, for the sake of the following derivation of the electronic interaction potential, that within the diffuse layer exists the so-called Slipping plane. This plane separates the fluid that is dragged by the colloid from the fluid that flows around it. Experimentally speaking, this is where one measures the electrostatic potential of a surface, the ζ potential.

At thermodynamic equilibrium, the local ion concentration in the diffuse layer $c_i(\vec{r})$ of a given species should follow the Boltzmann distribution:

$$c_i(\vec{r}) = c_i^{\infty} \exp\left\{-\frac{z_i e \psi(\vec{r})}{k_{\mathrm{B}}\Theta}\right\} \ , \tag{1.7}$$

where $c_{i,\infty}$ is the bulk or infinitely far from surfaces ion concentration, z_i is the valence of the ion, e is the elementary charge, and $\psi(\vec{r})$ is the electrostatic potential at position $\vec{r}$, which satisfies the Poisson equation:

$$\nabla^2\psi(\vec{r}) = -\frac{1}{\varepsilon}\sum_i z_i e c_i(\vec{r}) \ , \tag{1.8}$$

where ε is the permittivity of the fluid. Substituting the Boltzmann distribution for $c_i(\vec{r})$ into the Poisson equation yields the Poisson-Boltzmann equation, which is a non-linear partial differential equation for $\psi(\vec{r})$.

Thus, combining the above expressions, we get:

$$\nabla^2\psi(\vec{r}) + \frac{e}{\varepsilon}\sum_i z_i c_i^\infty \exp\left\{\frac{-z_i e\psi(\vec{r})}{k_\mathrm{B}\Theta}\right\} = 0 \ . \tag{1.9}$$

We only draw our interest to sufficiently dilute solutions in terms of electrolyte concentration $[NaCl]$. In that case, we may Taylor expand the exponential term to obtain:

$$\nabla^2\psi(\vec{r}) + \frac{e}{\varepsilon}\sum_i z_i c_i^\infty \left(1 - \frac{z_i e\psi(\vec{r})}{k_\mathrm{B}\Theta}\right) = 0 \ . \tag{1.10}$$

Since the fluid is electrically neutral,

$$\sum_i z_i c_i^\infty = 0 \ . \tag{1.11}$$

Thus, Eq. 1.10 becomes:

$$\nabla^2\psi(\vec{r}) = \left[\sum_i \frac{z_i^2 e^2 c_i^\infty}{\varepsilon k_\mathrm{B}\Theta}\right]\psi(\vec{r}) \tag{1.12}$$

$$= \frac{1}{l_\mathrm{D}}\psi(\vec{r}) \ , \tag{1.13}$$

where l_D is Debye's length,

$$l_\mathrm{D} = \sqrt{\sum_i \frac{\varepsilon k_\mathrm{B}\Theta}{z_i^2 e^2 c_i^\infty}} \ . \tag{1.14}$$

The above expression can be used to compute the Debye length in a given solution. For the case of a monovalent salt in water, we may keep in mind the simple relation $l_\mathrm{D} = 0.304\sqrt{C}^{-1}$, with C the salt concentration in solution.

Now, we compute the electrostatic repulsion potential between a sphere and a plane. To do so, it is convenient to consider the plane as a sphere of infinite radius, thus reducing the problem to a sphere-sphere interaction. The Debye-Huckel expression for the electrostatic potential around a single charged sphere is given in spherical coordinates by:

$$\frac{1}{r^2}\left[\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r)}{\partial r}\right)\right] = \frac{1}{l_{\mathrm{D}}^2}\psi(r)\ . \tag{1.15}$$

This equation has general solution:

$$\psi(r) = K_1\frac{\exp(\frac{r}{l_{\mathrm{D}}})}{r} + K_2\frac{\exp(\frac{-r}{l_{\mathrm{D}}})}{r} \tag{1.16}$$

Taking into account that $\lim_{r\to\infty}\psi(r) = 0$, we identify $K_1 = 0$. Thus, we can write

$$\psi(r) = K_2\frac{\exp(\frac{-r}{l_{\mathrm{D}}})}{r}\ . \tag{1.17}$$

We now are left with the identification of K_2. To do so, we use Gauss's flux theorem, and we finally get:

$$\psi(r) = \frac{\sigma a^2}{\varepsilon}\frac{\exp(\frac{a}{l_{\mathrm{D}}})}{1+\frac{a}{l_{\mathrm{D}}}}\frac{\exp\left(\frac{-r}{l_{\mathrm{D}}}\right)}{r}\ , \tag{1.18}$$

where σ is the charge density at the surface of the sphere, a is its radius.

Now that we have the electrostatic potential around a single sphere, we can suppose that the potential between two spheres is simply additive by approximation:

$$\psi_{\mathrm{tot}}(r) = \psi_1(r) + \psi_2(r)\ , \tag{1.19}$$

and with $r = a_1 + a_2 + z$ along sphere line of centers and keeping in mind cylindrical symmetry, the total electrostatic interaction potential along the z-direction becomes:

$$U_{\mathrm{tot}}(z) = \frac{4\pi}{\varepsilon}\left(\frac{\sigma_1 a_1^2}{1+l_{\mathrm{D}}^{-1}a_1}\right)\left(\frac{\sigma_2 a_2^2}{1+l_{\mathrm{D}}^{-1}a_2}\right)\frac{\exp(-l_{\mathrm{D}}^{-1}z)}{a_1+a_2+z}\ . \tag{1.20}$$

We now have the expression of the interaction potential between two charged spheres. As stated above, we may set one of the two radii to infinity and work with this limit to obtain the interaction potential between a sphere and a plane, as:

1

$$\frac{U_{\mathrm{elec}}}{k_{\mathrm{B}}\Theta} = B\mathrm{e}^{-\frac{z}{l_{\mathrm{D}}}} \ , \tag{1.21}$$

where

$$B = \frac{4\pi}{k_{\mathrm{B}}\Theta\varepsilon}\left(\frac{\sigma a^2}{1 + l_{\mathrm{D}}^{-1}a}\right) l_{\mathrm{D}}\sigma_{\mathrm{wall}} \ . \tag{1.22}$$

We furthermore neglect the attractive Van der Waals interactions, which in our cases of study only play a role within a few nanometers, whereas the repulsive interactions take place up to a few tens of nanometers.

Equilibrium distribution

Finally, we may summarize the total potential to which the colloid is subjected when immersed in water above a charged plane as:

$$U(z) = U_{\mathrm{g}} + U_{\mathrm{elec}} \ , \tag{1.23}$$

or, more in detail,

$$\frac{U(z)}{k_{\mathrm{B}}\Theta} = B\mathrm{e}^{\frac{-z}{l_{\mathrm{D}}}} + \frac{z}{l_{\mathrm{B}}}; \qquad \forall z > 0 \ , \tag{1.24}$$

as we would do not allow the colloid to cross the boundary.

As I discussed in the Introduction, the probability to find the colloid at position z given its diffusive behaviour, the gravitational and electrostatic potentials, should follow the Gibbs-Boltzmann equilibrium distribution:

$$P_{\mathrm{eq}} = \frac{1}{Z}\exp\left(-\frac{U(z)}{k_{\mathrm{B}}\Theta}\right) \ . \tag{1.25}$$

This distribution should be normalised, hence the $\frac{1}{Z}$ factor. It is particularly useful to keep the shape of it in mind when designing a simulation, as depending on the effect one wants to probe (very dense suspension, monolayer, or inversely, virtually unbounded colloids in the z-plane), we may play with the parameters to get the desired behaviour.

Fokker-Plank equation with multiplicative noise

Now, in the case of an inhomogeneous and anisotropic diffusion coefficient, the noise in our diffusion equation cannot simply be treated as a Gaussian white noise anymore. Indeed, because its amplitude depends on the diffusivity, and because the diffusivity is neither homogeneous nor isotropic anymore, we need to include a so-called multiplicative noise in the Langevin equation. This raises the question of the integration of the multiplicative noise, and the intent of the following derivation is to demonstrate the care that must be taken to ensure we obey detailed balance and do not introduce unphysicality by honest mistake.

We begin by taking the following Langevin equation (See the Introduction for its formal presentation), but modify the second term in the right-hand side to include a space-dependent amplitude:

$$\mathrm{d}X_t = u(X_t)\mathrm{d}t + g(X_t)\mathrm{d}B_t \ , \tag{1.26}$$

where, as a reminder, $u(X_t)$ is the drift from external forces acting upon the colloid, $g(X_t)$ is the space-dependent noise amplitude, and $\mathrm{d}B_t$ is a Wiener process, such that $\mathrm{d}B_t = \zeta(s)\mathrm{d}s$, $\langle \mathrm{d}B_t \rangle = 0$ and $\mathrm{d}B_t^2 = \mathrm{d}t$, which can be seen as the integral of a white noise: it is nowhere differentiable. Integrating Eq. 1.26 over an infinitesimal time step Δt,

$$X_{t+\Delta t} = X_t + \int_t^{t+\Delta t} u(X_s)\mathrm{d}s + \int_t^{t+\Delta t} g(X_s)\mathrm{d}B_s \ . \tag{1.27}$$

It is straightforward to simply integrate the first term of the right-hand side owing to the smoothness of $\mathrm{d}s$ using simple methods such as a forward Euler scheme [11]. However, the second term of the right-hand side is known as a stochastic integral, where B_s is a process which has infinite variation. In that case, one cannot simply treat $\mathrm{d}B_s$ as a small and deterministic increment, which in turn forbids the use of simple Euler integration. Thus, in order to compute this second integral, we often express it as:

$$\int_t^{t+\Delta t} g(X_s)\mathrm{d}B_s \equiv g\left([1-\alpha]X_t + \alpha X_{t+\Delta t}\right)\Delta W(t) \ , \tag{1.28}$$

with $\Delta W(t)$ the Wiener increment, as:

$$\Delta W(t) = \int_t^{t+\Delta t} \mathrm{d}B_s \ , \tag{1.29}$$

where α encodes where in the time step we evaluate the integral. We need to choose a value for α between 0 (left endpoint) and 1 (right endpoint). Of course, conventions for such integrals exist [12]:

1. The Itô convention: $\alpha = 0$; $\int_t^{t+\Delta t} g(X_s)\mathrm{d}B_s \simeq g(X_t)\Delta W$. The system reacts to the noise, there is no feedback within a single timestep. Mathematically, the SDE becomes a *martingale* [1] This means that the process's evolution depends strictly on the previous time step, and not on the history before this time step. However, no matter how convenient and practical this convention may be, it also entrails a modification, or rather correction to the chain rule of calculus when changing variables. We discuss this further down, as we will be using this convention for our SDE.

2. The Stratonovich convention: $\alpha = 1/2$; $\int_t^{t+\Delta t} g(X_s)\mathrm{d}B_s \simeq g(\frac{X_t+X_{t+\Delta t}}{2})\Delta W$. Stratonovich proposes a way of circumventing the complexities introduced by Itô's convention in the chain rule, breaking the martignale in the process. We now consider a noise which has a finite, non-zero correlation, but this correlation has a characteristic time that is very small compared to the discretisation time step. I will not detail it as much, as I have not really used this convention and am much less aware of its capabilities and pitfalls.

Mathematically, it is always possible to identify a Stratonovich process to an Itô process with equivalent solutions. Therefore, it is simply a matter of convenience to choose one over the other. There are, of course, subtleties; and I invite the interested reader to read more on the matter in L. Smith's website [13] from which I liberally adapted some content within this thesis. We'll, for the sake of the development, keep α as a variable for the time being.

We begin by taking Eq. 1.27 to order $\Delta t^{1/2}$:

$$X_{t+\Delta t} = X_t + g(X_t)\Delta W(t) \tag{1.30}$$

We can then conveniently develop the $X_{t+\Delta t} - X_t$ term in Eq. 1.28

[1] I was unaware of the reason for using Itô convention over other counterparts prior to writing the manuscript. However, upon reading literature, I understood how practical working with martingales can be. A martingale has a bounded mean, and, given that we know its history up to time $t' < t$, its expected value at t must be $\mathbb{E}\left[M_t | \{M_s, s \leq t'\}\right] = M_{t'}$. In my words, this means that the expected value of a martingale at a given time is the last previously known expected value. The name, however, comes from gambling, a field for which I have exactly zero interest to explore.

$$\int_t^{t+\Delta t} g(X_s)\mathrm{d}B_s\,\mathrm{d}s = g\big(X_t + \alpha[X_{t+\Delta t} - X_t]\big)\,\Delta W(t) \tag{1.31}$$
$$\approx g\big(X_t + \alpha g(X_t)\Delta W(t)\big)\,\Delta W(t) \tag{1.32}$$
$$\approx g(X_t)\Delta W(t) + \alpha g(X_t)g'(X_t)[\Delta W(t)]^2 \tag{1.33}$$

to order Δt. This in turn gives:

$$X_{t+\Delta t} = X_t + \big[u(X_t) + \alpha g(X_t)g'(X_t)\big]\,\Delta t + g(X_t)\Delta W(t)\ . \tag{1.34}$$

Notice then that the integration of a multiplicative noise brings a correction on the term which depends on Δt, a.k.a. the *drift* term. We standardise this expression to easily express the correction to the drift in our equation in order to simulate a trajectory. To do so, we invoke the Fokker-Plank equation yet again:

$$\frac{\partial P(X_t,t)}{\partial t} = -\frac{\partial}{\partial X_t}K_1(X_t)P(X_t,t) + \frac{1}{2}\frac{\partial^2}{\partial X_t^2}K_2(X_t)P(X_t,t)\ , \tag{1.35}$$

where K_1 and K_2 respectively are the first and second moments, which we define as:

$$\begin{aligned} K_1(X_t) &= \lim_{\Delta t\to 0}\frac{\Delta X_t}{\Delta t} \\ &= u(X_t) + \alpha g(X_t)g'(X_t) \end{aligned} \tag{1.36}$$

for the mean, and

$$K_2(X_t) = \lim_{\Delta t\to 0}\frac{(\Delta X_t)^2}{\Delta t} \tag{1.37}$$
$$= g^2(X_t) \tag{1.38}$$

for the standard deviation. Thus, we can rewrite Eq. 1.35 in those new terms:

$$\frac{\partial P(X_t,t)}{\partial t} = -\frac{\partial}{\partial X_t}\left\{u(X_t)+\alpha g(X_t)g'(X_t)\right\}P(X_t,t)+\frac{1}{2}\frac{\partial^2}{\partial X_t^2}\left\{g(X_t)^2\right\}P(X_t,t)\ . \tag{1.39}$$

Derivation of the stochastic drift term

We would now like to apply this knowledge to compute the new stochastic drift term one must add in the dt order to achieve physicality in our system. We begin again with the Langevin equation of a colloid sedimenting above a plane:

$$\mathrm{d}z = \frac{F(z)}{\gamma_\perp(z)}\mathrm{d}t + \sqrt{2D_\perp(z)}\mathrm{d}B_t\ . \tag{1.40}$$

In that case, taking Eq. 1.36, we express the mean velocity:

$$\left\langle \frac{\Delta z}{\Delta t}\right\rangle \equiv \bar{v_d} = \frac{F(z)}{\gamma_\perp(z)} + \alpha\frac{\mathrm{d}D_\perp(z)}{\mathrm{d}z}\ . \tag{1.41}$$

where $F(z)/\gamma_\perp(z)$ is our term of interest, and it is worth noting it contains the complexities of the following derivations and is, in no way, trivial and simply deriving from potentials.

We begin by taking the steady-state of Eq. 1.39 and differentiating with respect to time. By definition,

$$\frac{\mathrm{d}}{\mathrm{d}z}\left\{-u(z)-\alpha g(z)g'(z)+\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}g(z)^2\right\}P_{\mathrm{steady-state}}(z) = 0\ . \tag{1.42}$$

with $u(z)=\frac{F}{\gamma_\perp}$ and, because $\lim_{z\to\infty}P_{\mathrm{steady\ state}}(z)=0$ and $\lim_{z\to\infty}\frac{\mathrm{d}P_{\mathrm{steady\ state}}(z)}{\mathrm{d}z}=0$,

$$(-u(z)-\alpha g(z)g'(z))P_{\mathrm{steady-state}}(z)+\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}\left\{g(z)^2P_{\mathrm{steady-state}}(z)\right\} = 0\ . \tag{1.43}$$

We may now substitute $A(z)=g^2(z)P_{steady-state}(z)$, which yields:

$$\frac{\mathrm{d}A(z)}{\mathrm{d}z} - 2\frac{u(z)+\alpha g(z)g'(z)}{g^2(z)}A(z) = 0\ , \tag{1.44}$$

which integrates as:

$$\ln\left(A(z)\right) = \ln\left(g^2(z)\right) + \ln\left(P_{\text{steady state}}\right) \qquad = \int \frac{u(z') + \alpha g'(z')g(z')}{g^2(z')/2} \mathrm{d}z' \; . \tag{1.45}$$

And thus,

$$P_{steady-state}(z) = \frac{1}{g^2(z)} \exp\left\{ 2 \int_0^z \frac{u(z') + \alpha g(z')g'(z')}{g^2(z')} \mathrm{d}z' \right\} \; , \tag{1.46}$$

We note that we may express $\ln\left(g^2(z)\right) = \int \frac{2g'(z')g(z')}{g^2(z')}\mathrm{d}z'$, and

$$U_{steady-state}(z) = -k_\mathrm{B}\Theta \int \frac{u(z') + (\alpha - 1)g(z')g'(z')}{g^2(z')/2} \mathrm{d}z' \; . \tag{1.47}$$

Remembering that $F_{\text{steady state}} = -\frac{\mathrm{d}U_{\text{steady state}}}{\mathrm{d}z}$, we get:

$$\frac{F_{\text{steady state}}}{\gamma_\perp(z)} = \frac{F(z)}{\gamma_\perp(z)} + (\alpha - 1)g(z)g'(z) \; . \tag{1.48}$$

From the Lanvegin equation Eq. 1.40, we note that $g(z) = \sqrt{2D_\perp(z)}$, and $F(z) = \frac{u(z)}{\gamma_\perp}$, and we also keep in mind that $D_\perp = \frac{k_\mathrm{B}\Theta}{\gamma_\perp}$. With the Itô convention $\alpha = 0$, we may now write the total drift of Eq. 1.51:

$$\bar{v_\mathrm{d}} = \frac{1}{\gamma_\perp(z)} \frac{\mathrm{d}U_{\text{steady−state}}(z)}{\mathrm{d}z} + \frac{\mathrm{d}D_\perp(z)}{\mathrm{d}z} \; . \tag{1.49}$$

By identification, we then recover the deterministic drift velocity emerging from external forces:

$$v_\mathrm{d} = \frac{k_\mathrm{B}\Theta}{\gamma_\perp(z)} \left[-\frac{1}{l_\mathrm{D}} B \exp\left(-\frac{z}{l_\mathrm{D}} \right) + \frac{1}{l_\mathrm{B}} \right] \; , \tag{1.50}$$

whereas the stochastic drift, in the Padé approximation, reads:

$$v_{\text{stochastic drift}} = 2D_0 a \frac{2a^2 + 12az + 21z^2}{(2a^2 + 9az + z^2)^2} \; . \tag{1.51}$$

In the latter parts of this chapter, I use an algorithm that is specifically made to handle Brownian motion in confined geometries. It is worth keeping in mind the present derivation holds for this algorithm, which is why I deemed it relevant to be included in this thesis.

Simple Langevin simulation of a confined colloid

A very usual task in my group, thanks to the theses produced and defended by Maxime Lavaud [14] and Elodie Millan [15], is to simulate the system I just spent a few pages describing throughly. Thought not strictly part of my doctoral work (it was more of a Master's project under Maxime and Elodie's supervisions), I briefly include it here both to pay my respects to them and to show what can be done with minimal computations and basic code. I invite the interested reader to check their theses for ready-to-run, Cython [2]-compiled codes [16], and I provide minimal versions here (TODO : make a little jupyterlab for the reader to play around online. Think about whether to make one file per fig or a whole working example with simu and amalysis in the same file like Ryker did for libmobility in zaragoza.).

$$x_i = x_{i-1} + \sqrt{2D_{\parallel}(z_{i-1})\Delta t} w_{x_i} \ , \tag{1.52}$$

$$z_i = z_{i-1} + \bar{v_d}(z_{i-1})\Delta t + \sqrt{2D_{\perp}(z_{i-1})} w_{z_i} \ , \tag{1.53}$$

where w_{x_i} and w_{z_i} are two independent Gaussian white noises of zero mean and Δt variance, and where $\bar{v_d}(z)$ is the average drift velocity of the colloid at position z as computed in Eq. 1.51, which includes both deterministic and stochastic contributions as per equation 1.41.

We then carefully choose a time step such that ... Reference the introductory part. Write this part when I have rewritten the code. to do : -explanation of code, figure of traj, peq, MSD.

1.2.2 Colloid-colloid lubrication theory

Colloids displace fluid when they diffuse. Intuitively, it is very easy to understand that, if you were to the place two colloids or more in an otherwise unbounded fluid, their dynamics should couple in *some way*. Naturally, this question has been investigated through and throughout already, and I start by presenting the problem in terms of mobility $\mathbf{M}$ instead of diffusion. I explain how Oseen formalism [17] can be used to

[2] Cython refers both to a compiler and a programming language that is based on Pyrex. Owing to its high level of abstraction, Cython is extremely easy to write, maintain and comprehend, all while reaching the optimization capabilities of pure C, C++ or Fortran code. When I developed algorithms myself from scratch, I mostly favoured Python, and implemented computationally heavier loops in Cython.

probe at this coupling. I'll, however, mitigate the accuracy of the method, and present a second way of computing more accurately (and more expensively) the same quantities.

The mobility problem

We begin with N colloids submerged in an unbounded bulk fluid. We work at low Reynolds number (Re $\ll 1$) and neglect inertial effects. In that case, the Stokes equation become linear

$$-\nabla p(\boldsymbol{r}) + \eta \nabla^2 \boldsymbol{u}(\boldsymbol{r}) = -\boldsymbol{F}\delta(\boldsymbol{r}) \ , \tag{1.54}$$

$$\nabla \cdot \boldsymbol{u} = 0 \tag{1.55}$$

where ∇p is the pressure gradient in the fluid, that is to say, the force density due to pressure differences in the fluid; $\mu \nabla^2 \vec{u}$ is the viscous term that controls how the fluid diffuses momentum through itself with μ the fluid viscosity and $\vec{u}$ the fluid velocity field; $F\delta(\vec{r})$ is the point force at origin with δ the Kronecker function; and the second equation denotes fluid incompressibility.

The velocity vector can be written as a product of the mobility tensor $\mathbf{M}$ and the force vector $\mathbf{f}$:

$$\mathbf{v} = \mathbf{M} \cdot \mathbf{f} \ , \tag{1.56}$$

where the mobility matrix $\mathbf{M}$ is of dimension $6N \times 6N$ and has the structure:

$$\mathbf{M} = \begin{bmatrix} \boldsymbol{\mu}_{11}^{tt} & \cdots & \boldsymbol{\mu}_{1N}^{tt} & \boldsymbol{\mu}_{11}^{tr} & \cdots & \boldsymbol{\mu}_{1N}^{tr} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\mu}_{N1}^{tt} & \cdots & \boldsymbol{\mu}_{NN}^{tt} & \boldsymbol{\mu}_{N1}^{tr} & \cdots & \boldsymbol{\mu}_{NN}^{tr} \\ \boldsymbol{\mu}_{11}^{rt} & \cdots & \boldsymbol{\mu}_{1N}^{rt} & \boldsymbol{\mu}_{11}^{rr} & \cdots & \boldsymbol{\mu}_{1N}^{rr} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\mu}_{N1}^{rt} & \cdots & \boldsymbol{\mu}_{NN}^{rt} & \boldsymbol{\mu}_{N1}^{rr} & \cdots & \boldsymbol{\mu}_{NN}^{rr} \ . \end{bmatrix} \tag{1.57}$$

and $\mathbf{f} = \mathbf{F}_1, \ldots, \mathbf{F}_N, \mathbf{T}_1, \ldots, \mathbf{T}_N$ with $\mathbf{F}$ the forces, and $\mathbf{T}$ the torques. Subsequently,

$$\boldsymbol{v}_i = \sum_j \left(\mu^{\mathbf{tt}}_{\mathbf{ij}} \cdot \mathbf{F}_{\mathbf{j}} + \mu^{\mathbf{tr}}_{\mathbf{ij}} \cdot \mathbf{T}_{\mathbf{j}}\right) \ , \tag{1.58}$$

$$\boldsymbol{w}_i = \sum_j \left(\mu^{\mathbf{rt}}_{\mathbf{ij}} \cdot \mathbf{F}_{\mathbf{j}} + \mu^{\mathbf{rr}}_{\mathbf{ij}} \cdot \mathbf{T}_{\mathbf{j}}\right) \ , \tag{1.59}$$

with $\boldsymbol{v}_i$ the translational and $\boldsymbol{w}_i$ rotational velocities of the colloid indexed i. It is worth mentioning that in our problems, the mobility matrix must be symmetric positive definite [18], which greatly simplifies its structure.

The Oseen tensor

We consider the case of a colloid moving in an infinite and quiescent fluid. We aim to express the velocity field at any point in the fluid due to the displacement of this colloid. By effectively zooming out, we may approximate this colloid as a point force $\boldsymbol{F}_1$ acting upon the fluid at a given position $\boldsymbol{r_1}$.

In that case, the velocity field $\boldsymbol{u(r)}$ is written:

$$\boldsymbol{u}(\boldsymbol{r}) = \boldsymbol{G}(\boldsymbol{r} - \boldsymbol{r}_1) \cdot \boldsymbol{F} \tag{1.60}$$

where $\boldsymbol{G}$ is the Oseen tensor. Because $\boldsymbol{u}(\boldsymbol{r})$, is the fundamental solution of the Stokes equation (for a point force, that is), it is referred to as a "Stokeslet" [1], and the Oseen tensor $\boldsymbol{G}((r) - \boldsymbol{r}_1)$ is the Green's function to the Stokes equations. We do not take into account rotation of the colloids, that we consider as perfect spheres. We however keep track of their orientation anyway, for the sake of the derivation and later on to check numerical accuracy. With this, we may now express the flow field generated for a continuous force in space as the superposition of Green's functions:

$$\boldsymbol{u}(\boldsymbol{r}) = \int \boldsymbol{G}(\boldsymbol{r} - \boldsymbol{r'}) \cdot \boldsymbol{f}(\boldsymbol{r'}) \mathrm{d}^3 r' \ . \tag{1.61}$$

Solving the Stokes equation, and taking into account symmetries ($\frac{1}{4\pi}$ factor) and incompressibility ($\frac{1}{2}$ factor), we then obtain:

$$\boldsymbol{G}(\boldsymbol{r}) = \frac{1}{8\pi\eta r}(\mathbf{I} + \hat{\boldsymbol{r}}\hat{\boldsymbol{r}}) \ . \tag{1.62}$$

Now, if we solve for two colloids at positions $\boldsymbol{x}$ and $\boldsymbol{r_j}$ in interaction in the fluid, we simply obtain:

$$\boldsymbol{G}(\boldsymbol{x}-\boldsymbol{r}_j) = \frac{1}{8\pi\eta}\left(\frac{1}{|\boldsymbol{x}-\boldsymbol{r_j}|} + \frac{(\boldsymbol{x}-\boldsymbol{r_j})(\boldsymbol{x}-\boldsymbol{r_j})}{|\boldsymbol{x}-\boldsymbol{r}_j|^3}\right) \tag{1.63}$$

Moving on to the observable of interest for us, the diffusion tensor, we may express it in its full form as:

$$\langle \Delta r_{i,\alpha}(\Delta t)\Delta r_{j,\beta}(\Delta t)\rangle = 2D_{i\alpha,j\beta}\Delta t \ , \tag{1.64}$$

where the α and β components correspond respectively a sphere moving along the spheres lines of centers ($\|$) or perpendicularly to their lines of center ($\perp$) (Fig. 1.2.2) [6]. Taking Eq. 1.60, we may identify the $\boldsymbol{G}$ tensor as a mobility tensor. In that case, a diffusion tensor between two colloids can be expressed with $\boldsymbol{G}$ as:

$$D_{i\alpha,j\beta} = k_{\mathrm{B}}\Theta G_{\alpha\beta}(\vec{r}_1 - \vec{r}_2) \ . \tag{1.65}$$

Thus, the diagonal terms of the tensor are simply the self-diffusion of a colloid, whereas the off-diagonal blocks are the actual cross-diffusion we are interested in. By diagonalising the tensor, we retrieve

$$\frac{D_{\perp}^{C,R}}{2D_0} = 1 \pm \frac{3}{4}\frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right) \ , \tag{1.66}$$

$$\frac{D_{\|}^{C,R}}{2D_0} = 1 \pm \frac{3}{2}\frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right) \ , \tag{1.67}$$

where positive and negative corrections correspond respectively to the collective and relative modes of motion. As one may notice, hydrodynamic interactions as very long-range with respect to the colloid size and decay in $1/r$.

Now, as I discusse in this subsection, the Oseen approximation does not work for near-field interactions. Referring specifically to Eq. 1.67, our expressions are only valid down to $r \geq 10a$. Beyond that limit, the approximation becomes first incorrect, and then unphysical, with negative diffusion coefficients arising.

Luckily, because of its construction, the Oseen approximation is actually the first order of a more robust correction, which I detail in the next subsection.

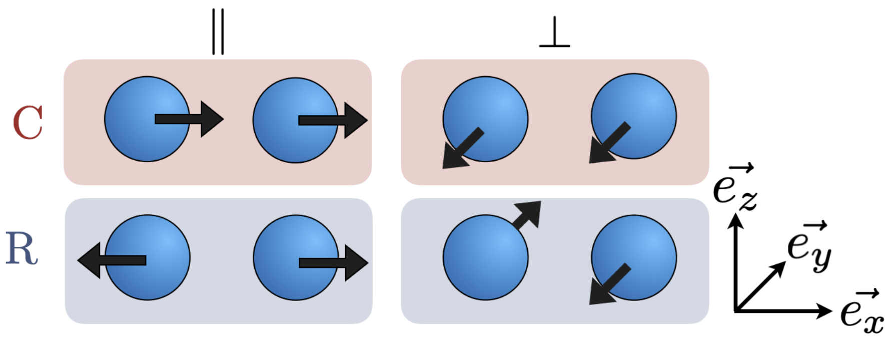

Figure 1.5: Diagram of the different modes of motion of interest for two Brownian colloids. For the sake of simplicity and continuity with the rest of the chapter, we do not consider the vertical direction.

The Rotne-Prager-Yamakawa tensor

To remedy the issues of the Oseen approximations mentioned above, we introduce the Rotne-Prager-Yamakawa (RPY) tensor, which incorporates finite size particles into the hydrodynamic interactions, at the pairwise level. This correction is, therefore, not a full N-body solution (which is anyway not solvable in a closed, analytic form, and would be a numerical nightmare to compute efficiently). I would put it, simply, as a finite-size correction to the Oseen approximation. Explicitly, for the two-colloid case,

$$\boldsymbol{\mu}_{12}^{\mathrm{RPY}}(\boldsymbol{r}) = \frac{1}{8\pi\eta\boldsymbol{r}}\left[\left(1+\frac{2a^2}{3r^2}\right)\mathbf{I}+\left(1-\frac{2a^2}{r^2}\frac{\boldsymbol{rr}}{r^2}\right)\right] . \tag{1.68}$$

This expression also ensures positive-definiteness of the mobility tensor. We will see, later down, that it is used in a modified form to account for the existence of boundaries in the fluid.

Should I detail this more guys ? I feel like, compared to the rest of my derivations, it is a little bit hollow, but on the other side I have not exactly derived it ever, nor have I felt the need to.

1.2.3 Coupling colloid-wall and colloid-colloid lubrication theories

Naturally, the next step is to couple both lubrication effects together. Luckily, such study has already been carried out both theoretically and experimentally by Dufresne et al. [6]. In this article, the authors discuss the modification on colloid cross-diffusion due to wall and colloid-colloid-induced lubrication effects. The following setup, depicted

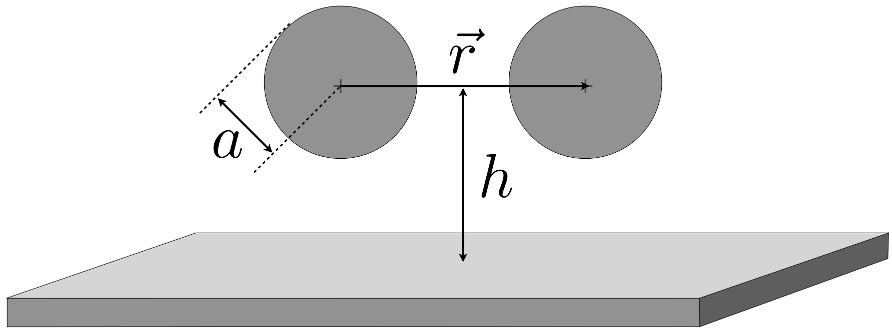

Figure 1.6: Experimental setup of Dufresne et al. [6] for the study of two colloids diffusing over a plane wall.

in Figure 1.2.3, is used: two colloids of radius $a = 0.485 \pm 0.025\mu$m (Duke Scientific lot 21024) are submerged in a water-salt solution of 2mM [NaCl] and viscosity $\eta = 0.817$cP. The water-salt layer is $140 \pm 2\mu$m thick, and sealed between a microscope slide and a No. 1 coverslip, with the system fully sealed with UV-cured epoxy to prevent evaporation of the fluid and fluid flows in the bulk. The system is maintained at $T = 29.00 \pm 0.01$°C. In these conditions, the colloid self-diffusion is found to be $D_0 = \frac{k_B \Theta}{6\pi\eta a} = 0.550 \pm 0.028 \mu\text{m}^2/\text{s}$.

The authors then trap the two colloids with a single optical tweezer, by alternating the position of the focal plane at the two positions of interest at a frequency of 200Hz using a galvanometer-driven mirror. Then, by blocking the beam for $t = 83$ms, they retrieve the sphere positions over five video frames only to ensure minimal out-of-focal displacement. From there on, authors carry out experiments over a grid of starting positions in (r, h) space, with r the center-to-center distance between colloids and $2\mu\text{m} < r < 10\mu\text{m}$, and h the colloid center-to-wall distance, with $1\mu\text{m} < h < 30\mu\text{m}$. By reactivating the trapping, they can repeat the experiment with perfect reproducibility for each starting position of interest in their system.

As discussed in the previous sections, colloids in a same bath should see their motion coupled through hydrodynamic interactions, but the coupling to the wall itself should also modify their expected diffusion. To investigate these coupling effects, the authors decompose the trajectories of the colloids into cooperative and relative motions, respectively $\vec{\rho} = \vec{r_1} + \vec{r_2}$ and $\vec{r} = \vec{r_1} - \vec{r_2}$ along ($\|$) and perpendicular ($\perp$) directions to the line of centers of the colloids (Fig. **??**). From there, they fill displacement histograms for each mode of motion at each of the five time steps taken into consideration and retrieve the effective diffusion coefficients from the standard deviation of each histogram, as:

$$\langle \Delta r_{i,\alpha}(\tau) \Delta r_{j,\beta} \rangle = 2 D_{i\alpha, j\beta} \tau \ , \tag{1.69}$$

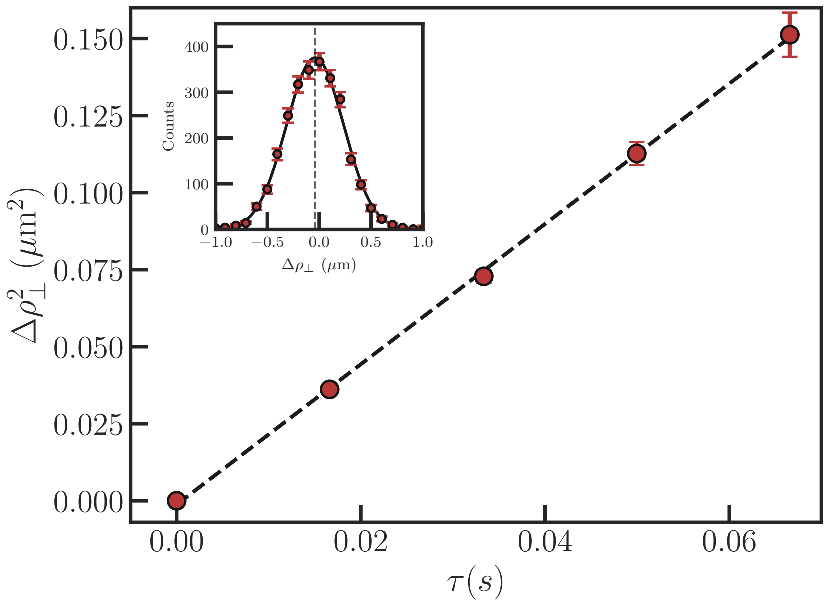

Figure 1.7: Adapted from [6]. Measured pair diffusion for the relative perpendicular mode of motion of the two colloids. The upper inset shows the histogram of all binned displacements for a given time step, for $h = 25.5 \pm 0.7\mu$m and $r = 7.00 \pm 0.25\mu$m. This process is repeated over all time steps and each histogram is fitted with a Gaussian curve to recover the root mean square displacement $\Delta\rho(r, h, \tau)$ to recover the MSD of the main plot. The effective cross-diffusion coefficient for the perpendicular mode is then extracted from the slope of the MSD.

with $i, j \in \{1, 2\}$ and $\alpha, \beta \in \{\parallel, \perp\}$, where τ is the elapsed time, and where $\langle . \rangle$ denotes an ensemble average.

A typical result of the experiment for a given set of (r, h) is shown in Figure 1.2.3. There, MSDs are extracted from displacement histograms at each of the five time lags into consideration. From there, authors can extract the effective cross-diffusion coefficient.

For comparisons purposes with their experimentally retrieved cross-diffusions in the four possible modes, the authors then compute by different means the theoretical expected value of the cross-diffusion coefficients.

Now, in the presence of a wall, one must take extra care to ensure no-slip boundary conditions at the wall as well as no fluid penetration through the wall itself. To do this, one may use the single-colloid method of images [19], where a Stokeslet (S), as Source Doublet (D) and a Stokeslet Doublet (SD) of opposite strength to the Stokeslet of interest at $\vec{r_j}$ above the wall is placed at equal distance below the wall, at $\vec{R_j} = \vec{r_j} - 2h\vec{e_z}$. The goal of the mirror Stokeslet is to cancel most of the normal penetration at the wall. The Source Doublet ensures no-slip at the wall, and the Stokeslet Doublet cancels remaining penetration. Then, the flow at the position of the image colloid is written:

$$G^W_{\alpha\beta}\left(\vec{x} - \vec{R_j}\right) = -G^S_{\alpha\beta}\left(\vec{x} - \vec{R_j}\right) + 2h^2 G^D_{\alpha\beta}\left(\vec{x} - \vec{R_j}\right) - 2hG^{SD}_{\alpha\beta}\left(\vec{x} - \vec{R_j}\right) \ , \quad (1.70)$$

with

$$G^D_{\alpha\beta}(\vec{x} - \vec{R_j}) = \frac{1}{8\pi\eta}\left(1 - 2\delta_{\beta z}\right)\frac{\partial}{\partial R_{j,\beta}}\left(\frac{(\vec{x} - \vec{R_j})_\alpha}{|\vec{x} - \vec{R_j}|^3}\right) \ , \quad (1.71)$$

$$G^{SD}_{\alpha\beta}(\vec{x} - \vec{R_j}) = (1 - 2\delta_{\beta z})\frac{\partial}{\partial R_{j,\beta}} G^S_{\alpha z}(\vec{x} - \vec{R_j}) \ . \quad (1.72)$$

which yields the two following modes of motion for the cross-diffusion coefficients

$$\frac{D_z}{D_0} = 1 - \frac{9}{8}\frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right) \ , \quad (1.73)$$

$$\frac{D_{xy}}{D_0} = 1 - \frac{9}{16}\frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right) \ . \quad (1.74)$$

One may notice the resemblance between the correction terms in the above expressions and the corrections in the Faxén formula from equation 1.3, which is not surprising as both corrections are due to the same wall-induced hydrodynamic effects. However, these expressions are only valid in the far-field regime, and fail to capture the lubrication effects in the near-field regime, which is precisely the regime of interest for our simulations.

The authors then propose an intermediate regime, first by naively superimposing the Oseen and Blake approximations as $D^{-1}_\psi(r,h) = D^{-1}_\psi(r) + \left[D^{-1}_{xy}(h) - D^{-1}_0\right]/2$ (in dash lines in Figure 1.2.3), which is a crude approximation which fails to capture the behaviour of the cross-diffusion for colloids below 50 radii from the plane. The authors then propose that the colloid must interact with its image, its neighbour, and its neighbour's image. In that case, the cross-diffusion is given by

$$\frac{D_{\perp}^{C,R}}{2D_0} = 1 - \frac{9}{16}\frac{a}{r} \pm \frac{3}{4}\frac{a}{r}\left(1 - \frac{1+\frac{3}{2}\xi}{(1+\xi)^{3/2}}\right) \tag{1.75}$$

$$\frac{D_{\parallel}^{C,R}}{2D_0} = 1 - \frac{9}{16}\frac{a}{r} \pm \frac{3}{2}\frac{a}{r}\left(1 - \frac{1+\xi+\frac{3}{2}\xi^2}{(1+\xi)^{5/2}}\right) \tag{1.76}$$

with $\xi = 4h^2/r^2$ and up to $\mathcal{O}\left(\frac{a^3}{r^3}\right)$ and $\mathcal{O}\left(\frac{a^3}{h^3}\right)$ (in solid lines in Figure 1.2.3).

Furthermore, as noted by the authors, in the very far-field regime, the Oseen approximations should be recovered numerically and experimentally, whereas they expect the above expressions to hold in the very near-field regime as the wall hinders coupling between colloid motions.

In the following parts, we confirm the above predictions numerically, but we push the study much deeper into the lubrication regimes to investigate its validity once we cannot ignore finite-size effects anymore in the lubrication approximations.

1.3 Simulating dense suspensions in close confinement

Naturally, researchers have been pondering on how to simulate dense Brownian suspensions for a few decades already [2]. However, the numerical complexity of such a method (and namely, how to efficiently solve for a multiplicative noise and, a fortiori, how to compute efficiently the square root of a diffusion or mobility tensor without computational cost blowups)...

Todo : finish this when i have had time to prepare a code, it will be clearer in my mind. add results and comparisons.

The lubrication-corrected version of the RigidMultiBlobsWall (RBW) algorithm [20] (here on GitHub, as stochastichydrotools/rigidmultiblobswall) aims at filling the gap [3] that is forbidden by the RPY approximation. As I mentioned earlier, it is computationally expensive to fully resolve the fluid and every boundary precisely. Therefore, the RBW algorithm proposes:

- In the far-field, to use the RPY approximation anyway, because it is exact in that regime;

[3] no pun intended

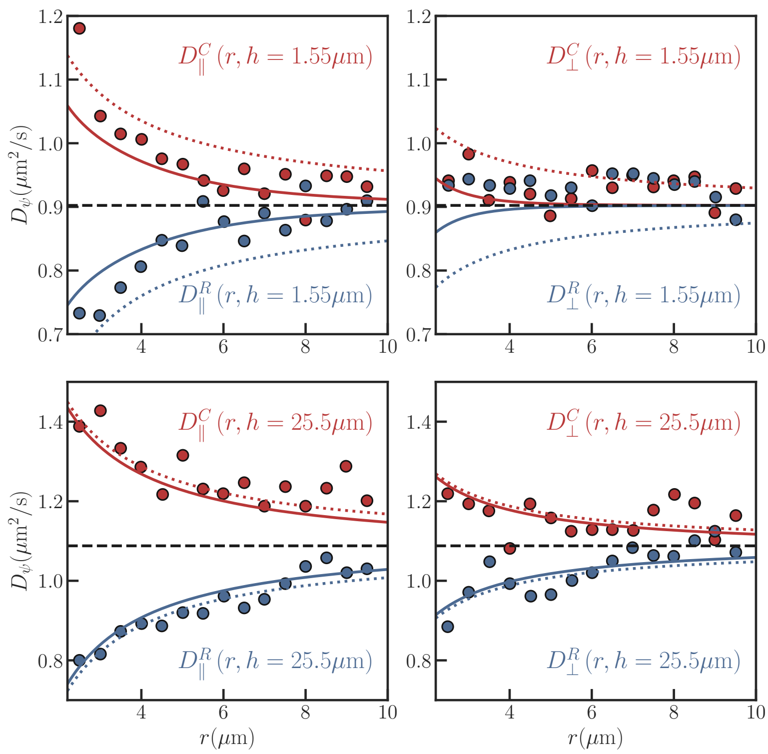

Figure 1.8: Adapted from [6]. Measured cross-diffusion coefficients for the relative perpendicular and parallel modes of motion of the two colloids, as a function of the center-to-center distance r between the colloids, for different values of the center-to-wall distance h. The dash lines correspond to the naive superposition of Oseen and Blake approximations, while the solid lines correspond to the more accurate approximation taking into account the interaction with the image colloid.

- In the near-field, to replace the RPY approximation by subtraction of its contribution and addition of asymptotical corrections to the mobility matrix.

In an effort of convenience, because the resistance matrix is additive, we introduce it as $\boldsymbol{R} = \boldsymbol{M}^{-1}$ the inverse of the mobility matrix.

We take again a suspension of N colloids, and enforce a no-slip boundary below on the $z = 0$ plane. Formally, we write the new, corrected mobility matrix of the whole system as:

$$\bar{\boldsymbol{M}} = [\boldsymbol{R} + \boldsymbol{R}_{\mathrm{lub}}^{\mathrm{sup}} - \boldsymbol{R}_{\mathrm{RPY}}^{\mathrm{sup}}]^{-1} , \tag{1.77}$$

where $\boldsymbol{R}$ is constructed from pairwise RPY approximations, $\boldsymbol{R}_{\mathrm{lub}}^{\mathrm{sup}}$ is the new, asymptotical corrections that are added to the resistance of nearby colloids, and $\boldsymbol{R}_{\mathrm{RPY}}^{\mathrm{sup}}$ is the erroneous RPY corrections that $\boldsymbol{R}$ includes due to the RPY not correcting the close-field appropriately. Todo : when I work again on the code, go over this, as right now i don't really remember it...

Chapter 2

Brownian Motion in Soft Confinement

2.1 Introduction

In many biological systems, one must not only take into account Brownian behaviours, but also carefully evaluate the effect of the system's or parts of the system's deformability on its dynamics. Indeed, if we take the example of a lipid bilayer, which is a soft and quasi-two-dimensional fluid, the thermal energy $k_{\mathrm{B}}\Theta$ of the system is sufficient to cause long-wavelength bending fluctuations of the cell membrane, thickness fluctuations, lipid movement within the plane itself, and so forth. Naturally, a neighbouring fluid's motion is then affected by these shape modifications, which can in turn affect the dynamics of a body evolving in that same bath. This coupling is not trivial.

The coupling between softness and viscous flows, which we will refer to as the elastohydrodynamic (EHD) coupling, has been extensively studied in the deterministic framework. We for example mention the emergence of lift forces near a deformable boundary at low Reynolds numbers, which have been first demonstrated theoretically [21] [22][23] [24] [25] and then experimentally [26] [27] [28] [29] [30].

The field of stochastic EHD is still fairly new. Molecular dynamics simulations have been conducted to investigate the anomalous diffusion of a micelle near a lipid bilayer [31]. Another recent study [32] demonstrated experimentally the emergence of a new, non-conservative force acting upon a Brownian soft colloid near a confining wall. However, we still to this day lack a precise theory to encompass quantitatively the long-time signatures of the coupling between softness, confinement and fluctuations.

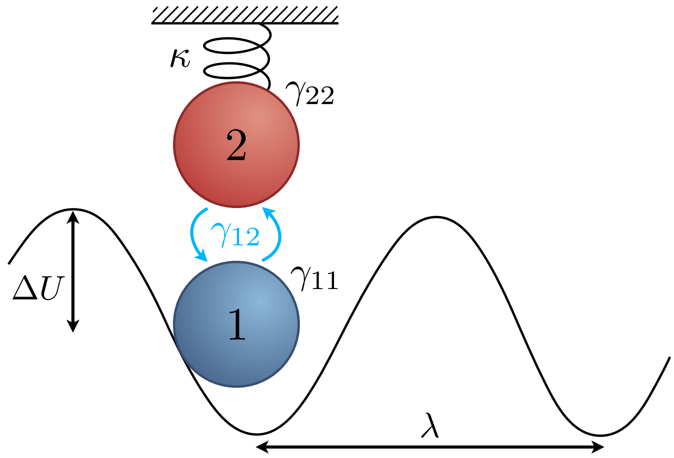

Figure 2.1: Representation of the system. A Brownian colloid of interest (blue) is trapped in a periodic potential of amplitude ΔU and spatial period λ. An elastic membrane is modelled locally as a second Brownian colloid (red) subject to a harmonic potential of inverse stiffness (or *compliance*) κ. This membrane is coupled to the colloid of interest through an indirect, hydrodynamic-like coupling represented by the coefficient γ_{12}, and both the colloid and the membrane feel their own self-friction γ_{11} and γ_{22}, respectively.

Because of the level of complexity of such a study, I present in this chapter a most simple model, with elementary ingredients, which can still qualitatively capture all the effects we are interested in. This model was published in *Communications Physics* in 2025 [33] and the full print and Supplementary Information can be found in appendix of this thesis. In the following sections, I present the theoretical aspects of the model, developed by David S. Dean. I develop the numerical model used to simulate this toy model. Finally, I show how numerics and theory compare.

2.2 Toy Model for soft Brownian Motion

2.2.1 Overview of the Toy Model

From here on, we study a purely one-dimensional case of a Brownian colloid coupled to an elastic and fluctuating membrane (Fig. 2.1). The Brownian colloid is represented by a first degree of freedom, q_1, and is submitted to a periodic trapping potential $\phi(q_1)$ of spatial period λ and amplitude ΔU. Throughout this chapter, the Brownian colloid's degree of freedom q_1 will be color-coded in blue, and the membrane's degree of freedom q_2 in red. We couple to this colloid the toy model version of the membrane. We represent the membrane with a colloid that is submitted in a harmonic potential of energy $\frac{q_2^2}{2\kappa}$, where κ is the compliance (or inverse stiffness). It represents the elasticity of the membrane.

This second colloid is tracked with the second degree of freedom, q_2.

Both the Brownian colloid and the membrane feel their own self-friction, respectively γ_{11} and γ_{22}. Furthermore, they interact through an indirect, hydrodynamic-like interaction γ_{12}. This interaction represents the effect of the fluid that is displaced by the motion of the colloid on the membrane, and *vice-versa*. All together, these terms form the friction matrix $\mathbf{\Gamma}$

$$\mathbf{\Gamma} = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{pmatrix} \tag{2.1}$$

which can be inverted to give the mobility matrix of the system, as $\mathbf{M} = \mathbf{\Gamma}^{-1}$. As per Lorentz's reciprocal theorem, both matrices must be positive definite, and, because their diagonal terms account for the self-friction and self-mobility respectively, $\mu_{11} > 0$ and $\mu_{22} > 0$.

We can write the total energy of the system as

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa} \ , \tag{2.2}$$

and we notice the absence of Hamiltonian interactions in the system. In that case, we work in the overdamped limit, and we write the coupled Langevin equations describing the evolutions of q_1 and q_2 as

$$\gamma_{11}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \gamma_{12}\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\phi'(q_1) + \eta_1(t) \ , \tag{2.3}$$

$$\gamma_{12}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \gamma_{22}\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\frac{q_2}{\kappa} + \eta_2(t) \ , \tag{2.4}$$

where η_i and η_j (with $j \in \{1, 2\}$) are Gaussian white noises of zero mean and correlation functions $\langle \eta_i(t)\eta_j(t')\rangle = 2k_{\mathrm{B}}T\delta(t - t')\gamma_{ij}$. The equilibrium probability density function of the system is then given by the Gibbs-Boltzmann distribution, as

$$P_{\mathrm{eq}} = \frac{\exp[-\beta H(q_1, q_2)]}{Z} \ . \tag{2.5}$$

2.3 Theoretical developments

The objective of the following developments is to compute the modification to the long-time dynamics of the Brownian colloid diffusing within the sinusoidal trap. The development I choose to present here is not the most rigorous, and corresponds to the very stiff ($\kappa \to 0$) limit of the elastic membrane. The interested reader can find a more rigorous argument based on multiscale analysis [34] [35] in the appendices of this thesis, alongside a second development for the very compliant limit ($\kappa \to \infty$).

We begin by refactoring Eqs. 2.3 and 2.4 in the deterministic case $\Theta = 0$

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_{11}\phi'(q_1) - \frac{\mu_{12}}{\kappa} q_2 \ , \tag{2.6}$$

$$\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\mu_{12}\phi'(q_1) - \frac{\mu_{22}}{\kappa} q_2 \ . \tag{2.7}$$

We then integrate to express q_2 as

$$q_2 = -\frac{\kappa\mu_{12}}{\mu_{22}} \left(1 + \frac{\kappa}{\mu_{22}}\frac{\mathrm{d}}{\mathrm{d}t}\right)^{-1} \phi'(q_1) \ . \tag{2.8}$$

In the small κ limit, we then expand the right-hand bracket term, taking into account the chain rule

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}} \left[\phi'(q_1) - \frac{\kappa}{\mu_{22}}\phi''(q_1)\frac{\mathrm{d}q_1}{\mathrm{d}t} + O(\tau_2^2)\right] \ . \tag{2.9}$$

Here, we note that τ_2 is the characteristic relaxation time of the harmonic trapping corresponding to the membrane elasticity, as $\tau_2 = \frac{\kappa}{\mu_{22}}$. Now, substituting this new expression for q_2 in Eq. 2.6, we obtain

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}}\right)\phi'(q_1(t)) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2}\phi''(q1(t))\frac{\mathrm{d}q_1}{\mathrm{d}t} \ , \tag{2.10}$$

which we can reexpress as

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_{\mathrm{e}}(q_1)\phi'(q_1) \ , \tag{2.11}$$

where μ_{e} is a dressed mobility with expression

$$\mu_{\mathrm{e}} = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22}\left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2}\right)} \ . \tag{2.12}$$

Remembering that $\mathbf{\Gamma} = \mathbf{M}^{-1}$,

$$\mu_{\mathrm{e}} = \frac{1}{\gamma_{11}\left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{11}^2}\right)} \ . \tag{2.13}$$

Now, we can avoid reexpressing the noise which should be applied to Eq. 2.11 for the thermal case by working directly with Fokker-Planck formalism. This formalism is introduced in its elementary form at the beginning of my thesis, and developed for more complex cases in Chapter 2. CHECK CHAPT NR AT THE END OF THE REDACTION HERE Doing such a thing is not rigorous, as I previously mentioned, because the effective stochastic process has not been derived. Eliminating fast variables in Eq. 2.11 typically introduces non-trivial corrections. However, since we assume separation of timescales, the effective slow process still permits the writing of a uniquely-defined Fokker-Plank equation.

This Fokker-Plank equation reads

$$\frac{\partial P_1(q_1,t)}{\partial t} = \frac{\partial}{\partial q_1}\mu_\mathrm{e}(q_1)\left[k_\mathrm{B}\Theta\frac{\partial P_1(q_1,t)}{\partial q_1} + \phi'(q_1)P_1(q_1,t)\right] , \tag{2.14}$$

and, noting that the time derivative of the probability density function $P_1(q_1,t)$ can be expressed in terms of probability flux J,

$$\frac{\partial P_1(q_1,t)}{\partial t} = -\frac{\partial J}{\partial q_1} . \tag{2.15}$$

This allows us to write

$$J = -D_\mathrm{e}(q_1)\frac{\partial}{\partial q_1}P_1(q_1,t) - \mu_\mathrm{e}\phi'(q_1)P_1(q_1,t) , \tag{2.16}$$

where $D_\mathrm{e} = k_\mathrm{B}\Theta$. The above equation can be rewritten

$$J = -D_\mathrm{e}(q_1)\exp\left[-\beta\phi(q_1)\right]\frac{\mathrm{d}}{\mathrm{d}q}\left\{\exp\left[\beta\phi(q_1)\right]P_1(q_1,t)\right\} \tag{2.17}$$

with $\beta = \frac{1}{k_\mathrm{B}\Theta}$.

Now, we tilt the potential by adding a small force F, which gives a new potential $U(q_1)$, as

$$U(q_1) = \phi(q_1) - Fq_1 . \tag{2.18}$$

Then, the current equation can be rewritten taking into account this new force, and gives

$$\frac{\mathrm{d}}{\mathrm{d}q_1}\{\exp[\beta U(q_1)]P(q_1,t)\} = -\frac{J(q_1,t)}{D_\mathrm{e}(q_1)}\exp[\beta U(q_1)] . \tag{2.19}$$

We can then integrate over a period λ, and, using periodicity, we retrieve

$$P_1(q_1,t) = \frac{J(q_1)\exp[-\beta U(q_1)]}{1-\exp(-\beta F\lambda)}\int_{q_1}^{q_1+\lambda}\frac{\exp[\beta U(q)]}{D_\mathrm{e}(q)}\mathrm{d}q . \tag{2.20}$$

By essence, $P_1(q_1, t)$ is normalised

$$\int_0^\lambda P_1(q_1)\mathrm{d}q_1 = 1 \;, \tag{2.21}$$

and thus we retrieve for J

$$J = \frac{1 - \exp[-\beta F\lambda]}{\int_0^\lambda \exp[-\beta U(q)]\mathrm{d}q \int_q^{q+\lambda} \frac{\exp[\beta U(q)]}{D_\mathrm{e}(q)}\mathrm{d}q} \;. \tag{2.22}$$

In the small force limit $F \to 0$,

$$1 - \exp(-\beta F\lambda) \simeq \beta F\lambda \;, \tag{2.23}$$

which, to leading order in F, gives

$$J = \frac{\beta F\lambda}{\int_0^\lambda \exp[-\beta\phi(q_1)]\mathrm{d}q_1 \int_0^\lambda \frac{\exp[\beta\phi(q_1)]}{D_\mathrm{e}(q_1)}\mathrm{d}q_1} \;. \tag{2.24}$$

Finally, we note that the drift velocity should read $v = J(q_1)\lambda$, and that the effective diffusion coefficient over λ should be

$$D^* = \frac{k_\mathrm{B}\Theta v}{F} = \frac{\lambda^2}{\int_0^\lambda \exp[-\beta\phi(q_1)]\mathrm{d}q_1 \int_0^\lambda \frac{\exp(\beta\phi(q_1))}{D_\mathrm{e}(q_1)}\mathrm{d}q_1} \;, \tag{2.25}$$

which is the generalised Lifson-Jackson formula [36].

Finally, we replace $D_\mathrm{e}(q_1)$ in the expression

$$D^* = \frac{k_\mathrm{B}\Theta\lambda^2}{\gamma_{11}\int_0^\lambda \mathrm{d}q \exp[\beta\phi(q)]\left(1 - \frac{\kappa\gamma_{12}^2\phi'^2(q)}{k_\mathrm{B}\Theta\gamma_{11}^2}\right)\int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)]} \;. \tag{2.26}$$

And, setting $\phi(q_1) = \Delta U\phi(\frac{q_1}{\lambda})$, where ΔU is the amplitude of the periodic trap

$$D^* = \frac{k_\mathrm{B}\Theta}{\gamma_{11}\int_0^1 \mathrm{d}p \exp[\beta\Delta U\psi(p)]\left(1 - \frac{\kappa\gamma_{12}^2\Delta U^2\psi'^2(p)}{k_\mathrm{B}\Theta\gamma_{11}^2}\right)\int_0^1 \mathrm{d}p \exp[-\beta\Delta U\psi(p)]} \;. \tag{2.27}$$

Which is the equation we use to compare with numerical results.

In the $\kappa \to 0$ limit, the above expression can be approximated as

$$D^* = D^*(0)(1+\alpha) \;, \tag{2.28}$$

with

$$\alpha = \beta\epsilon \frac{\lambda^2\gamma_{12}^2}{\gamma_{11}^2} \frac{\int_0^\lambda \mathrm{d}q_1 \phi'(q_1)^2 \exp[\beta\phi(q_1)]}{\int_0^\lambda \mathrm{d}q_1 \exp[-\beta\phi(q_1)]} \ , \tag{2.29}$$

where we define $\epsilon = \frac{k_B\Theta}{\lambda^2}$ a dimensionless compliance. As $\alpha > 0$, the coupling of the membrane to the Brownian colloid always increases its long-time diffusion constant.

Furthermore, in the very compliant limit $\kappa \to \infty$, D^* becomes

$$D^* = k_B\Theta \frac{\mu_{11}(\mu_{11}\mu_{22} - \mu 12^2)}{\lambda^{-2}Z_+Z_-(\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O\left(\frac{1}{\sqrt(\epsilon)}\right) \ . \tag{2.30}$$

We notice, in that regime, the absence of a κ or ϵ dependence.

2.4 Numerical modelling

2.4.1 Discretisation

My main mission in this project is to simulate the above Eqs. 2.3 and 2.4 efficiently. To do so, I discretise these equations using a simple forward Euler scheme (which was introduced in the beginning of this thesis). The equations read

$$q_1(t+\Delta t) - q_1(t) = -\mu_{11}\phi'(q_1(t))\Delta t - \frac{\mu_{12}}{\kappa} q_2(t)\Delta t + \xi_1\sqrt{2k_BT\Delta t} \ , \tag{2.31}$$

$$q_2(t+\Delta t) - q_2(t) = -\frac{\mu_{22}}{\kappa} q_2(t)\Delta t - \mu_{12}\phi'(q_1(t))\Delta t + \xi_2\sqrt{2k_BT\Delta t} \ , \tag{2.32}$$

where ξ_i is a zero-mean Gaussian random variable, as

$$\langle \xi_i \xi_j \rangle = \mu_{ij} \ . \tag{2.33}$$

We choose to work in terms of mobility instead of friction matrix elements for convenience in the upcoming developments. They are, however, strictly equivalent.

Furthermore, in the rest of this study, we choose $\phi(q_1)$ to be sinusoidal, as $\phi(q_1) = \Delta U \sin\left(\frac{2\pi q_1}{\lambda}\right)$, with ΔU the amplitude of the sinusoidal trap.

2.4.2 Time step selection

In any discretisation process, it is essential to choose the appropriate time step. This time step must be small enough to avoid a blowup from error propagation (again, see the beginning of this thesis for more details on how the Euler method propagates errors), but big enough to remain advantageous. An arbitrarily small timestep prohibits the probing of very long time dynamics due to computation time.

To select a correct time step for a simulation, I typically first identify all the smaller time scales in the system I want to simulate. In the present study, I identify two time scales : the local relaxation time within a local minimum of the sinusoidal trap, and the relaxation time of the harmonic trap.

Near a local minimum, we approximate the sinusoidal trap as harmonic. In that case,

$$\phi(q_1) = \frac{1}{2} k_{\text{eff}} q_1^2 \ , \tag{2.34}$$

where k_{eff} is an effective stiffness, which we identify as the local curvature of the potential as

$$\begin{aligned} k_{\text{eff}} &= |\phi''(0)| & (2.35) \\ &= \Delta U \left(\frac{2\pi}{\lambda}\right)^2 \ , & (2.36) \end{aligned}$$

and we can finally identify this first timescale as

$$\tau_1 = (\mu_{11} k_{\text{eff}})^{-1} = \frac{\lambda^2}{4\pi^2 \Delta U \mu_{11}} \ . \tag{2.37}$$

The harmonic trap is much simpler, as it requires no approximations, and we find its characteristic timescale

$$\tau_2 = \frac{\kappa}{\mu_{22}} \ . \tag{2.38}$$

Then, I choose a time step which satisfies

$$\Delta t = \frac{1}{100} \max \tau_1, \tau_2 \ . \tag{2.39}$$

We also need to define an upper bound to the simulation time. We must always remain mindful of the computational cost of our simulations, as time and energy both are scarce resources. Thus, we can define a timescale τ^* which depends on the long-time effective diffusion coefficient $D^*(\kappa)$ defined in Eq. 2.27

$$\tau^*(\kappa) = \frac{\lambda^2}{D^*(\kappa)} \ . \tag{2.40}$$

I choose to run my simulations for a time T_{sim} which is much larger than τ^*, as $T_{\text{sim}} = 100\tau^*$. This allows me to probe the long-time dynamics of the system, and to extract the effective diffusion coefficient from the mean-squared displacement of the Brownian colloid. This, in turn, sets the computational cost of a simulation, which I can quantify using the total number of iterations in the time loop N as

$$N = \left\lfloor \frac{T_{\text{sim}}}{\Delta t} \right\rfloor = \left\lfloor \frac{100\tau^*}{\Delta t} \right\rfloor . \tag{2.41}$$

2.5 Results

The interested reader can find a minimal version of a trajectory generation code here. This simple code allows the user to tune parameters and test for themself their effect on the dynamics of the system. Most of the figures presented in this section are generated using this simpler script, in which I left the seed fixed to ensure proper reproducibility.

2.5.1 Trajectories

I first generate trajectories of the two coupled Brownian colloids using Eqs. 2.31 and 2.32, for a given set of parameters. A sample of a single trajectory is shown in Fig. 2.2. We can see that the membrane fluctuates around its equilibrium position in $q_2 = 0$, whereas the Brownian colloid is trapped at short times within a local minimum, and escapes through random thermal fluctuations at longer times to explore space. This is a first verification that the simulations behave as expected. Indeed, it is not unusual for Brownian simulations to blowup in the case of an unfortuitous dimensionalisation of the system, and this issue is almost always imputable to a too-large time step. We have already discussed this matter in the beginning of the manuscript.

We verify the behaviour of the system at equilibrium by sampling the probability density function of q_1 and q_2 from the generated trajectories, and comparing them to the theoretical prediction from the Gibbs-Boltzmann distribution (Eq. 2.5). This is shown in Fig. 2.3. We can see that the sampled probability density functions (histograms) and theoretical prediction from the Gibbs-Boltzmann distribution (solid lines, Eq. 2.5) are in perfect agreement, even for low statistics (a single binned trajectory of 10^8 points suffices to reach a good match between theory and numerics).

The sampled probability distribution of q_1 follows the expected periodic shape, whereas the one of q_2 follows the expected Gaussian shape predicted by Eq. 2.5. This is a second verification that the simulations are working as expected, and that we are correctly sampling the equilibrium distribution of the system.

However, at this stage, we have not probed the dynamics of the system. Thus, we still have no insight on the effect of a soft membrane on the diffusion of the Brownian

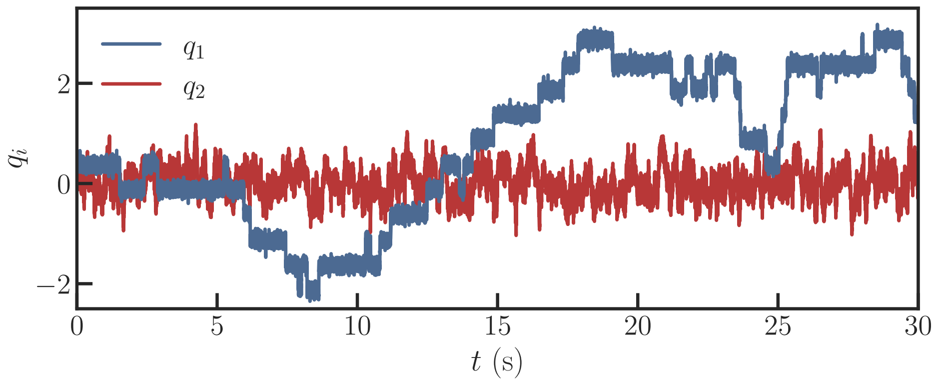

Figure 2.2: Trajectories of the Brownian colloid (blue) and the membrane (red) for $\kappa = 0.01$. The other parameters are fixed to $\Delta U = 2.7 k_\mathrm{B}\Theta$, $k_\mathrm{B}\Theta = 1.0$, $\lambda = 1.0$, $\gamma_{11} = 1.0$, $\gamma_{22} = 1.0$ and $\gamma_{12} = 0.8$, $\tau = 10^{-4}$. The membrane noticeably fluctuates around its equilibrium position in $q_2 = 0$, whereas the colloid is trapped at short times within a local minimum, and escapes through random thermal fluctuations at longer times to explore available space.

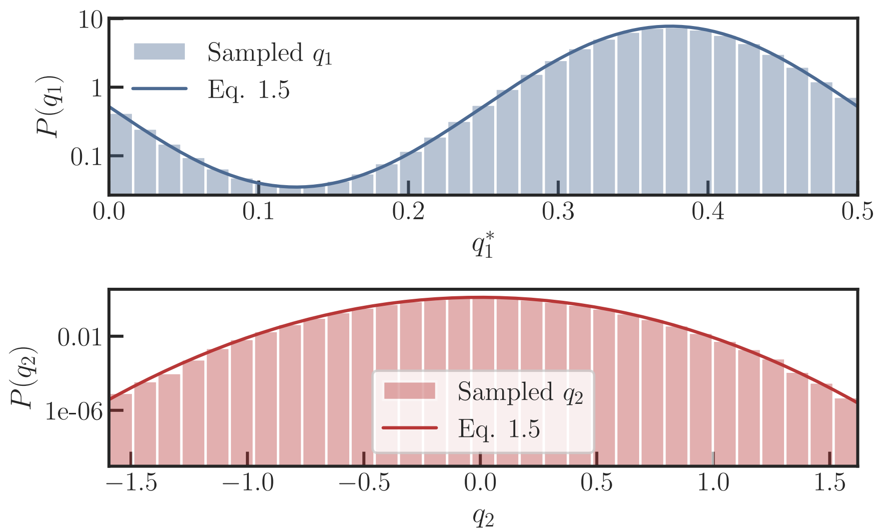

Figure 2.3: Equilibrium probability density function of q_1 (top, for $q_1^* = q_1 \mod \lambda$) and q_2 (bottom) for $\kappa = 0.01$. The other parameters are fixed to $\Delta U = 2.7 k_\mathrm{B}\Theta$, $k_\mathrm{B}\Theta = 1.0$, $\lambda = 1.0$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\tau = 10^{-4}$s, for a single trajectory of $N = 1 \times 10^7$ points. The sampled probability density functions (histograms) and theoretical prediction from the Gibbs-Boltzmann distribution (Eq. 2.5, solid lines) are in perfect agreement, even for a single trajectory of 10^8 points.

colloid. To do so, we will need to compute the MSD of q_1 and q_2 from the generated trajectories, and to extract the effective diffusion coefficient of q_1 from the long-time behaviour of its MSD. We will then be able to compare this effective diffusion coefficient to the theoretical prediction from Eq. 2.27.

2.5.2 Mean-Squared displacements

In the present case below, the MSD is computed directly from the generated trajectories, calculating the average of the squared displacements over all possible or given time lags. This means that the computer will store the whole trajectories q_1 and q_2 in memory. It will then loop over all possible time lags Δt, and for each of them, it will compute the average of the squared displacements $[q(t + \Delta t) - q(t)]^2$ over all possible time origins t. This technique is straightforward and sufficient for demonstration purposes.

Snippet 2.1: Direct computation of the MSD from trajectories

```
def MSD(x, lags):
    msd = np.zeros(len(lags)) # Always initialise you arrays if
        you know their size beforehand.

    for n, lag in enumerate(lags):
        if lag >= len(x):      # Avoid pitfalls.
            msd[n] = np.nan

        else:
            msd[n] = np.nanmean( # Ignore NaN values !
                (x[:-lag] - x[lag:])**2 # In words, NaN-dropped
                    mean of the square of displacements for timelag
                     lag of index n in the lag array.
            )

    return msd # Returns an array of size n with the MSD for each
        time lag in the lags array.
```

With a simple estimate looking at the code above, we can see that the computational cost of such a method scales as $O(N^2)$ at worst.

In our published work and in similar numerical studies [37], it is more convenient to first compute the *histograms of displacements* at given time lags. We can then simply extract the MSD from the width of the histogram of displacements, as

$$\mathrm{MSD}(\Delta t) = \sigma^2(\Delta t) \ , \tag{2.42}$$

where σ^2 is the variance of the histogram of displacements at time lag Δt.

Thought not computationally lighter, this is the technique I recommend for this type of statistical study for the reasons that follow. The histogram of displacements has the advantage of being a stackable quantity across iterations of a same simulation. On a rolling basis, as multiple trajectories are calculated, we simply add counts to the histogram of displacements. Furthermore, it is a much more general quantity than the MSD, as it allows for the computation of other moments of the distribution of displacements (skewness, kurtosis, etc.).

We can define a convergence criterion on the histogram of displacements as the relative distance between a "previous" histogram of cumulative displacements, and an "updated" histogram which contains all previous iteration plus the new additions from the latest simulated trajectories. When this relative distance is smaller than a given threshold for a sufficient number of iterations, we deem the simulation to have converged, and it automatically stops. This is a convenient way of saving resources and active working time, as it allows full automation of the whole process.

I define the convergence criterion as

$$\varepsilon = \frac{\sum_i |P_i^{\text{prev}} - P_i^{\text{updated}}|}{\sum_i P_i^{\text{prev}}} \ , \tag{2.43}$$

where P is the histogram of displacements, and i the index of the bin. I choose a threshold of $\varepsilon \leq 5 \times 10^{-3}$ for more than 3 iterations of 10 stacked trajectory statistics. This means that, when the relative distance between the "previous" and "updated" histograms of displacements is smaller than 5×10^{-3} for more than 3 iterations of 10 stacked trajectories, we consider the simulation to have converged, and it automatically stops.

The results of a typical MSD calculation for q_1 and q_2 are shown in Figs. 2.4 and 2.5. We can see that the MSD of q_1 (Fig. 2.4) is at first diffusive with a Δt exponent of 1 and a diffusion coefficient D_{m} which is directly proportional to the self-mobility of the Brownian colloid as

$$D_{\text{m}} = k_{\text{B}} \Theta \mu_{11} \ . \tag{2.44}$$

It then falls into a slow crossover to reach a second diffusive regime of effective diffusivity D^* as defined in Eqs. 2.27 and 2.30 for the very rigid and very compliant cases. The long-time diffusion coefficient is smaller than the short-time diffusion coefficient, which is a signature of the trapping effect of the sinusoidal potential.

The MSD of q_2 (Fig. 2.5) is also diffusive at short times with a diffusion coefficient D_2

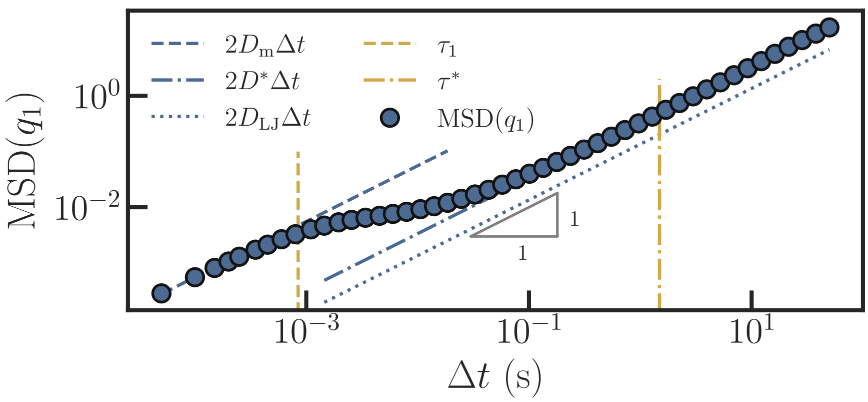

Figure 2.4: Mean-Squared Displacement of q_1 for $\kappa = 0.01$, $\Delta U = 2.7 k_{\mathrm{B}}\Theta$, $k_{\mathrm{B}}\Theta = 1.0$, $\lambda = 0.5$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\tau = 10^{-4}$s, for a single trajectory of $N = 1 \times 10^{8}$ points. At small timelags $\Delta t < \tau_1$, the Brownian colloid presents a diffusion constant D_{m} (Eq. 2.44, in blue dashes) which corresponds to a bulk, free diffusivity inside a local minimum of the potential $\phi(q_1)$. It then slowly crosses over to a second diffusive regime for $\Delta t > \tau^{*}$ with diffusion constant D^{*} (Eq. 2.30, in blue dashdot). The dotted blue line indicates the expected long-time diffusion coefficient in the very rigid limit $\kappa \to 0$ (Eq. 2.27 for $\kappa = 0$), which is smaller, as expected.

which is directly proportional to the self-mobility of the membrane as

$$D_2 = k_{\mathrm{B}}\Theta\mu_{22} \ , \tag{2.45}$$

and then crosses over to a confined regime at long times $\Delta t > \tau_2$ due to the harmonic trapping of the membrane. We can predict the value of the long-time MSD plateau of q_2 as $\mathrm{MSD}(q_2) = 2k_{\mathrm{B}}\Theta\kappa$, which we verify numerically.

2.5.3 Quantification of diffusion enhancement

We can finally infer all diffusion coefficients by performing affine fits of the MSDs in the diffusive regimes, remembering that in the diffusive regime

$$\mathrm{MSD}(\Delta t) = 2D\Delta t \ . \tag{2.46}$$

In log scale, a diffusive regime corresponds to a linear regime with a time exponent of 1 as $\log(\mathrm{MSD}) = \log(2D) + \log(\Delta t)$, and the diffusion coefficient can be directly inferred from the intercept of the affine fit to the MSD axis at the origin of Δt.

For a given set of parameters, we can then compare the effect of a varying compliance

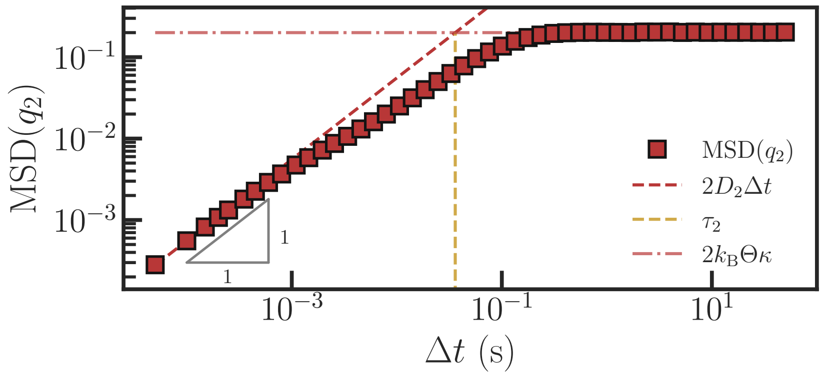

Figure 2.5: Mean-Squared Displacement of q_2 for $\kappa = 0.01$, $\Delta U = 2.7k_\mathrm{B}\Theta$, $k_\mathrm{B}\Theta = 1.0$, $\lambda = 0.5$, $\gamma_{11} = 1$, $\gamma_{22} = 1$, $\gamma_{12} = 0.8$, $\tau = 10^{-4}$s, for a single trajectory of $N = 1 \times 10^8$ points. At small timelags $\Delta t < \tau_2$, the membrane presents a diffusion constant D_2 (Eq. 2.45, in red dashes) which corresponds to a bulk, free diffusivity inside the minimum of the harmonic potential. It then slowly crosses over to a second diffusive regime for $\Delta t > \tau^2$ with constant MSD. The dotted red line indicates the expected plateau value for the long-time MSD at $2k_\mathrm{B}\Theta\kappa$.

of a soft membrane on the long-time diffusion coefficient of a Brownian colloid trapped nearby in a sinusoidal potential (Fig. 2.6).

We define an augmentation factor $\frac{\Delta D^*(\epsilon)}{D^*}$ as

$$\frac{\Delta D^*(\epsilon)}{D^*} = \frac{D^*(\epsilon) - D^*(0)}{D^*(0)} \,, \tag{2.47}$$

to conveniently compare the coupled to uncoupled case. We notice that numerical results seem in perfect agreement with the expected theoretical augmentation factors in the asymptotical regions $\epsilon \to 0$ and $\epsilon \to \infty$. Interestingly, one may think of ϵ as a ratio of timescales. At very small ϵ, the membrane is extremely stiff, and q_2 becomes the fast mode. At very large ϵ, on the contrary, it comparatively becomes the slow mode.

Around $\epsilon = O(1)$, however, the timescale separation is lost. The particle coordinate and the elastic degree of freedom now relax on comparable timescales, and thus, neither Lifson-Jackson expression is valid.

To access this crossover regime, we solve the full two-dimensional Kubo problem [1][38]. All the relevant developments and details on the numerical solution of the Kubo equation

[1]for which the interested reader can find the FlexPDE code in appendix of this manuscript.

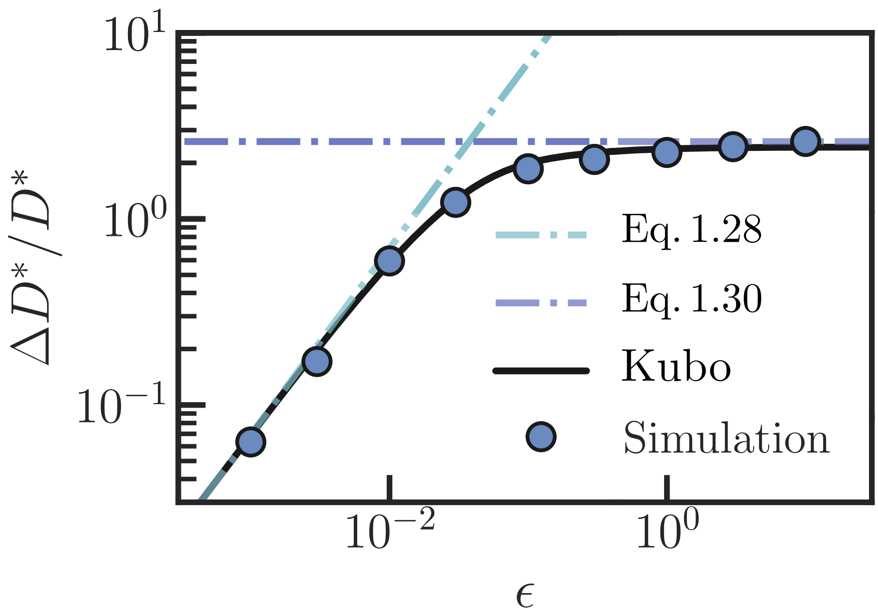

Figure 2.6: Augmentation factor $\frac{\Delta D^*(\epsilon)}{D^*}$ as a function of the coupling parameter ϵ for $\Delta U = 3.0 k_\mathrm{B}\Theta$, $k_\mathrm{B}\Theta = 1.0$, $\lambda = 1.0$, $\mu_{11} = 1$, $\mu_{22} = 1$, $\mu_{12} = 0.85$. The numerical results (blue dots) are in perfect agreement with the expected theoretical augmentation factors in the asymptotical regions $\epsilon \to 0$ and $\epsilon \to \infty$ (dash-dotted lines) of Eqs. as well as the crossover region, which is accessed by solving the full two-dimensional Kubo problem (solid black line).

can be found in the appendix of this thesis. I will not go into details here, as I did not personally work on that matter.

This numerical solution does not require assuming either $\epsilon \ll 1$ or $\epsilon \gg 1$, and therefore gives the crossover between the two asymptotic regimes. It is, however, much more tedious to estimate due to the computational cost of solving.

Furthermore, we may input to the model biological values for the parameters. Let us consider a colloid of radius $a = 100$nm at a distance $d = 10$nm from the membrane of elastic modulus $E = 1$kPa. The equivalent compliance of the membrane can be found using Hertz's model to give $\kappa = (E\sqrt{ad})^{-1}$. The system is immersed in a fluid of viscosity $\eta = 10^{-3}$Pa.s, at room temperature $T = 300$K. The Brownian colloid is subject to a periodic potential of amplitude $\Delta U = 4k_{\mathrm{B}}\Theta$ and spatial period $\lambda = a = 100$nm.

To recover mobility values, we assume $\mu_{11} = mu_{22} = (6\pi\eta a)^{-1}$, and assume Oseen's mobility expression for μ_{12} as $\mu_{12} = (4\ pi\eta r)^{-1}$, with r the center-to-center distance between the Brownian colloid and the colloid representing the membrane, as $r = 2a + d$.

With these values, the adimentionalised compliance ϵ falls exactly in the crossover region where the system is not considered very stiff or very compliant. Still, we may estimate the expected enhancement of the long-time diffusion coefficient by simply plotting these two asymptotes (Fig. 2.7) and recover at $\epsilon = 0.013$ an expected enhancement of about $\frac{\Delta D^*}{D^*} \approx 100\%$.

2.6 Conclusion and perspectives on the chapter

In this Chapter, we have exposed the novelty of the field of stochastic elastohydrodynamics. We have developed a toy model with elementary ingredients which encompasses qualitatively the physics of a Brownian colloid diffusing nearby a soft membrane. We have then developed a theoretical framework to predict the long-time diffusion coefficient of the Brownian colloid as a function of the compliance of the membrane, and we have implemented numerical simulations to verify these predictions. We have found that the coupling of the membrane to the Brownian colloid always increases its long-time diffusion constant given the framework of our model.

However, this systematic increase should be heavily modified by the introduction of more physical ingredients in the model. For instance, the attentive reader may have noticed our mobility matrix is not position-dependant, which hydrodynamically speaking would involve colloids quite far apart compared to their respective radii, rendering the coupling extremely weak. Furthermore, the system is so far in only one dimension, which

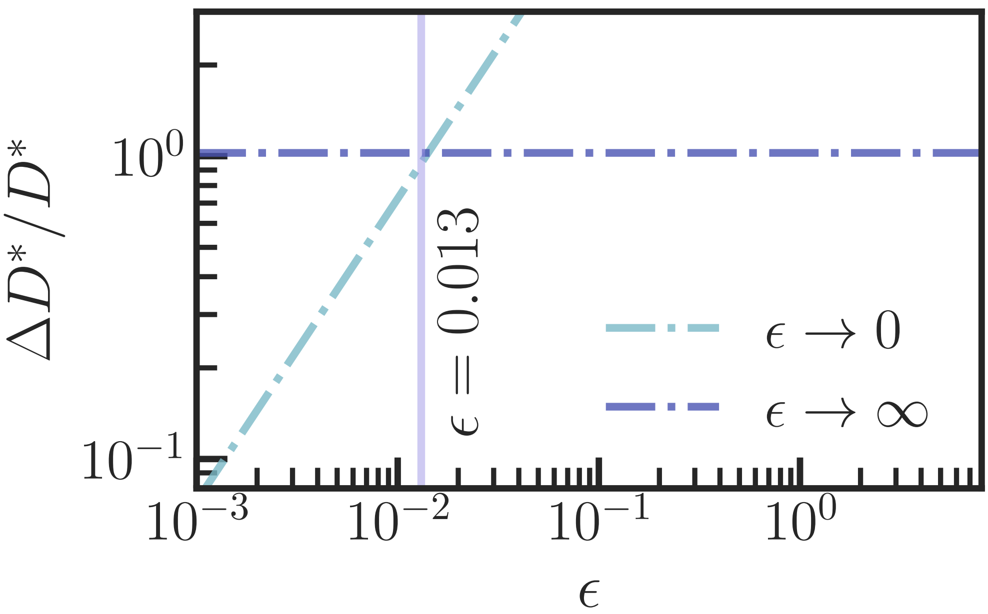

Figure 2.7: Estimation of the relative enhancement of the long-time diffusion coefficient of a Brownian colloid of radius $a = 100$nm in a periodic potential of amplitude $\Delta U = 4k_\mathrm{B}T$ and spatial period $\lambda = 100$nm at a distance $d = 10$nm from membrane of elastic modulus $E = 1$kPa or equivalent dimensionless compliance $\epsilon = 0.013$. The hydrodynamic constant μ_{12} is estimated using Oseen's model. The two asymptotes of the very rigid $\epsilon \to 0$ and very soft $\epsilon \to \infty$ are shown in dash-dotted cyan and blue lines respectively. For this system, the expected value of the relative enhancement of the long-time diffusion coefficient of the Brownian colloid is of about $\frac{\Delta D^*}{D^*} \approx 100\%$.

should modify quantitatively the expected enhancement on the diffusion coefficient of the Brownian colloid. We also work in overdamped regime, but aknowledge the interest of keeping mass in the system so as to probe at added mass effects which are currently being investigated within our group.

Nevertheless, the methodology presented in this Chapter as well as the numerical methods are solid and, as the time of writing of this manuscript, being extended to some cases mentioned above.

Chapter 3

Brownian motion in fluid flows

3.1 Introduction

Fluid flows are ubiquitous in many biological settings. The study of fluid dynamics provides insights on multiple ranges of applications. Oceanic currents and large-scale weather phenomena [39], the lift of an airplane foil [40], surfing [41] and drug biodisponibility all rely at least partly on a range of application of fluid dynamics. The latter is particularly relevant to my thesis, as it is a perfect example of the interplay between a fluid flow and Brownian motion for the diffusive transport of drugs in the bloodstream. In this Chapter, I will focus on the study of Brownian motion in fluid flows, and more specifically in viscous flows. This is a very common situation in microfluidic experiments.

3.1.1 The Reynolds number

We commonly use the Reynolds number R_e to predict the behaviour of a fluid. This number is a ratio between inertial and viscous forces in the fluid

$$R_\mathrm{e} = \frac{\rho U L}{\eta} \ , \tag{3.1}$$

where ρ is the fluid's density, U is a characteristic velocity, L is a characteristic length and η is the fluid's dynamic viscosity. Turbulence in a flow occurs at very high Reynolds number $R_\mathrm{e} \gg 1$, whereas a very small Reynolds number $R_\mathrm{e} \ll 1$ is considered laminar, with a layered and constant profile. A very classic example is that of cigarette smoke (Fig. 3.1). The smoke exists the cigarette's tip at very low velocity. By taking air density at room temperature $\rho_\mathrm{air} = 1.2\,\mathrm{kg.m}^{-3}$, viscosity at about $\eta_\mathrm{air} = 1.8\,\mathrm{Pa.s}$, and a smoke characteristic velocity $U_\mathrm{smoke} = 1\,\mathrm{cm.s}^{-1}$ and characteristic plume width $L_\mathrm{smoke} = 1\,\mathrm{mm}$,

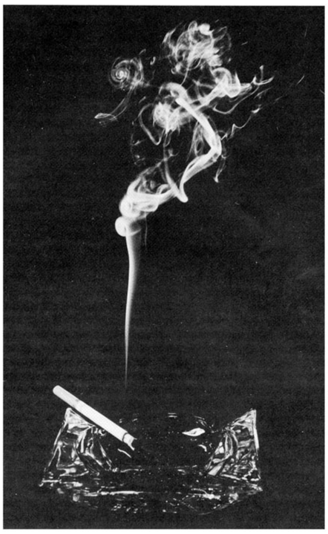

Figure 3.1: Smoke rising from a lit cigarette tip. At short distances from the tip, the smoke plume is narrow and the velocity low enough for the flow to be considered laminar. Higher up, the plume is wider and the velocity higher; the flow transitions to turbulent. Source: Physics Stack Exchange.

we can estimate the first regime at about $R_\mathrm{e} = 0.5$. Higher up, the smoke rises much faster and the plume is wider, with a characteristic speed of $U_\mathrm{smoke} = 10\,\mathrm{cm.s}^{-1}$ and a characteristic width of about $L_\mathrm{smoke} = 2\,\mathrm{cm}$, the ratio gives $R_\mathrm{e} \approx 100$. Perturbations are suppressed at low intensity due to viscous effects. As the speed rises, perturbations can no longer be damped, and the typical turbulences arise.

Most microfluidic experiments and simulations are conducted in the low-Reynolds-number regime. Despite the apparent simplicity of Stokes flows, this regime exhibits a wide range of rich phenomena, including non-trivial transport, mixing, particle migration, and microswimmer dynamics [42] [43].

In the framework of this thesis, I am interested in understanding flow effects on Brownian motion. Specifically, one question has already been very well investigated in the last century: *"What happens to a Brownian solute once it is advected by a low-Reynolds, viscous flow?"*

This Chapter first introduces the classical Taylor-Aris theory, developed to predict the dispersion of a Brownian solute under such flow. Then, I present an adjacent unpublished work carried out in collaboration with Joshua Mc. Graw, Blaise Delmotte and their respective groups, investigating hydrodynamic interactions in a dilute Brownian solute under flow advection.

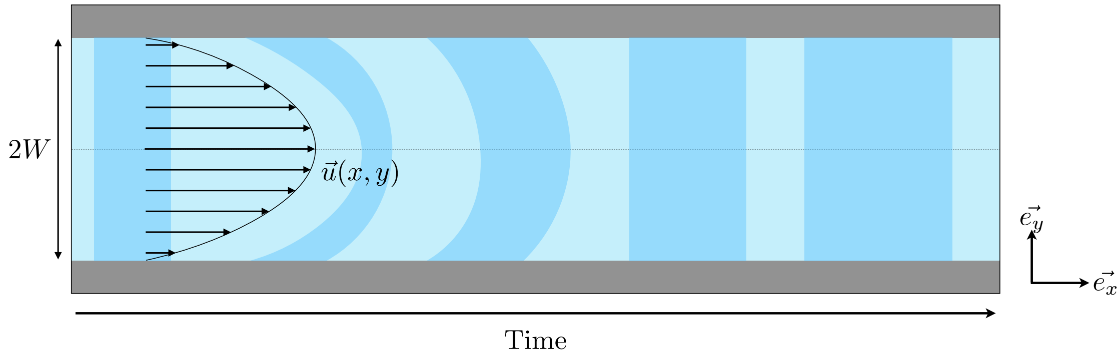

Figure 3.2: A plug of Brownian solute (darker blue) is injected in a microchannel of width $2W$. It is then advected by a viscous flow. Because of no-slip conditions at the walls (grey), the flow takes a parabolic, or Poiseuille, shape. At short times, the plug of solute takes the shape of the flow. However, because of thermal fluctuations, the plug diffuses across streamlines. This averages out the parabolic profile of the solute cross-section wise. At times larger than cross-channel diffusion time $t > \tau_{\mathrm{D}}$ (Eq. 3.4), the concentration profile of the solute resumes a homogeneous shape, with a larger width. This broadening of the width of the plug is quantified by the dispersion coefficient which is computed from Eq. 3.9.

3.2 Taylor-Aris dispersion

We turn our focus to the study of the evolution of a Brownian solute's concentration when advected by a viscous flow (Fig. 3.2).

We consider a microchannel of width $2W$. Fluid in that channel is advected by a unidirectional flow $u(y)\mathbf{e}_x$, and takes a characteristic Poiseuille (or parabolic shape) from the no-slip conditions at contact with the channel walls. A solute of concentration $c(x, y, t)$ and bulk diffusivity D_0 is injected in that microchannel. It obeys the advection-diffusion equation

$$\partial_t c + u(y)\partial_x c = D_0 \left(\partial_x^2 c + \partial_y^2 c\right) \ . \tag{3.2}$$

We decompose

$$c(x, y, t) = \bar{c}(x, t) + c'(x, y, t), \tag{3.3}$$

where $\bar{c}$ is the cross-sectional average and c' is a small transverse correction induced by the shear. At times larger than the transverse diffusion time

$$\tau_{\mathrm{D}} = \frac{W^2}{D_0} \ , \tag{3.4}$$

the concentration is equilibrated across the channel width. Solving for this correction and averaging over the transverse coordinate gives an effective one-dimensional advection-diffusion equation

$$\partial_t \bar{c} + \bar{u}\,\partial_x \bar{c} = D_{\mathrm{eff}} \partial_x^2 \bar{c} \ . \tag{3.5}$$

For a two-dimensional planar Poiseuille flow between plates separated by $2W$,

$$u(y) = \frac{3}{2}\bar{u}\left(1 - \frac{y^2}{W^2}\right) \ , \qquad -W < y < W, \tag{3.6}$$

the effective dispersion coefficient D_{eff} is then

$$D_{\mathrm{eff}} = D_0 + \alpha \frac{\bar{u}^2 W^2}{D_0} \ . \tag{3.7}$$

The α factor arises from channel geometry and varies with its configuration. From there, we can identify a new adimensionalised quantity, the Péclet number P_{e}

$$P_{\mathrm{e}} = \frac{\bar{u} W}{D_0} \ , \tag{3.8}$$

which compares advective to diffusive behaviour in the solute. Finally, we can write

$$D_{\mathrm{eff}} = D_0 \left(1 + \alpha P_{\mathrm{e}}^2\right) \ , \tag{3.9}$$

which is the classical Taylor-Aris formula [44] [45] [46] [47]. In simple words, this formula relates the dispersion of a Brownian solute in a microchannel of given width to the average fluid velocity for times beyond τ_{D}. This means that, for low enough Péclet numbers, the dispersion should be minimal, but that beyond $P_{\mathrm{e}}^2 > 1/\alpha$, dispersion should become dominant over classical free diffusion D_0 (Fig. 3.3). It is worth noting that, while I chose to speak in terms of concentration, the same phenomenon applies for a single colloid in a viscous flow.

Taylor-Aris dispersion is still a very much ongoing field of study. Researchers have incorporated to the original Taylor-Aris theory a range of sophistications, from time-varying flow profiles [48] [49] to particle size determination [50] [51] [52] [53], and even to active swimmers [54], which will be the focus of the next Chapter.

3.3 Hydrodynamic interactions in an advected solute

The reason I became personally interested in flow effects on Brownian motion in the first place is thanks to Masoodah Gunny (ESPCI) and her thesis supervisors Alexandre Vilquin and Joshua McGraw. Their group focuses among other thematiques on the prob-

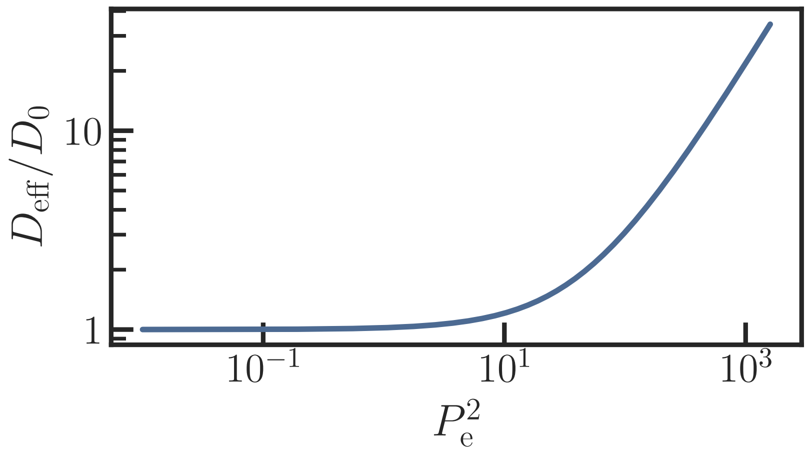

Figure 3.3: Predicted renormalised dispersion $D_{\mathrm{eff}}/D0$ against Péclet number squared P_{e}^2 computed from Eq. 3.9 with $\alpha = 1/48$, corresponding to the case of a cylindrical tube. At small P_{e}, $D_{\mathrm{eff}}/D0$ remains around 1, meaning that the dispersion effect due to advection is barely noticeable. It then crosses over around $P_{\mathrm{e}}^2 = 1$ to a second regime which is linear in P_{e}^2.

lem of Taylor-Aris dispersion near the solid-liquid interface. Specifically, they wonder in this project whether the volume fraction of a solute in a flow can influence the diffusion of this solute, even a very low density (this is akin to an ideal gas limit). Because this problem is highly nontrivial from experimental standpoint, their group together with Blaise Delmotte (LadHyX) suggested I conduct numerical simulations to replicate their system while in control of every parameter and interaction. I present here briefly their experimental protocol, their problematique and the results of our collaboration.

3.3.1 Experimental aspects

This part of the Chapter is largely based on Masoodah Gunny's PhD thesis manuscript [55].

Experiments are carried out with silica beads of radius $a = 55$nm in salted water of concentration [NaCl] $= 5.4 - 54\,\mathrm{mg.L}^{-1}$ (with corresponding Debye lengths $l_{\mathrm{D}} = 35 - 10\,\mathrm{nm}$). The solution is injected in a microfluidic channel setup of length $L = 8.8\,\mathrm{cm}$, width $W = 180\,\mu\mathrm{m}$ and thickness $H = 20\mu\mathrm{m}$. A pressure gradient ΔP is then applied between the entrance and exit of the channel to start the flowing. Very close to the wall, we can approximate the Poiseuille profile of the flow in the microchannel as a linear shear γ (Fig. 3.4), and we take the shear rate $\dot{\gamma}$ as $\dot{\gamma} = \frac{H}{2\eta}\Delta P$ with η the fluid's viscosity.

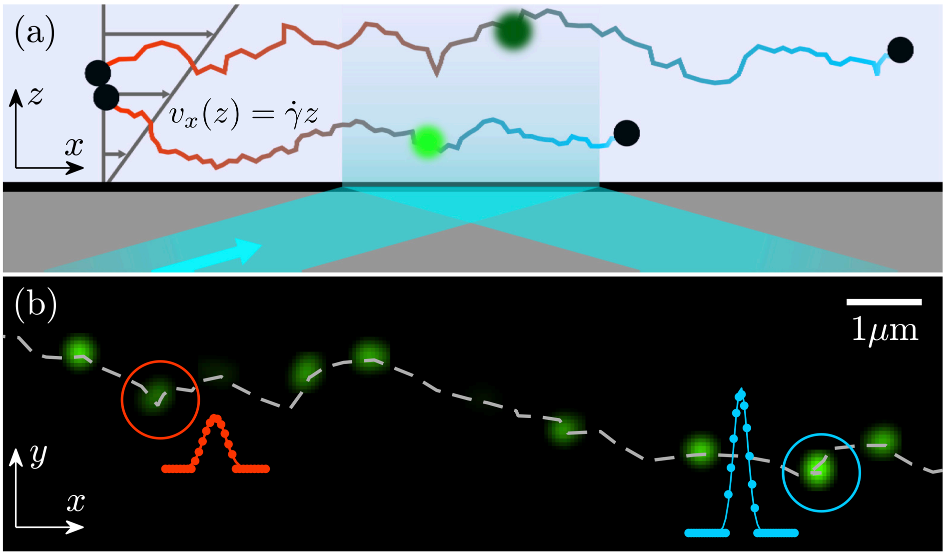

Figure 3.4: Scheme of the Taylor-dispersion TIRFM setup. (a) Two colloids of radius $a =$ 55 nm are advected by a shear flow of shear rate $\dot{\gamma}$. As they penetrate the evanescent field, they fluoresce and their positions in all three dimensions are recovered. (b) Experimental images for a single colloid. A succession of frames taken with lag time $\tau = 12.5$ ms is shown. Intensity profiles are represented in red and blue dots with Gaussian fits. The linked trajectory of the colloid is represented in a grey dashed line. Adapted from [56].

We are interested in individual colloid behaviour, and thus need an imaging technique that allows for the visualisation of individual colloids rather than a global concentration profile.

Imaging is done by Total Internal Reflection Microscopy (TIRFM) [57] (Fig. 3.5). TIRFM has the advantage of being a minimally disruptive, sub-wavelength imaging technique which still gives information on the three coordinate of a colloid's position within observation range. The principle of TIRFM relies on evanescence to collect information on the colloid's altitude. An incident laser beam hits the wall of the microchannel with incident angle θ. This angle must obey

$$\theta > \theta_c \,, \tag{3.10}$$

where θ_c is the critical angle above which no refraction is permitted, as

$$\theta_c = \sin^{-1}\left(\frac{n_1}{n_2}\right) , \tag{3.11}$$

with n_1 and n_2 the optical index of the water solution and the solid medium, respectively, and $n_1 < n_2$. Like this, refraction is prohibited, and almost all incident light is reflected. However, we know from Maxwell's equations that electromagnetic fields must remain continuous at the interface between the two media. Thus, the field must not end abruptly, but rather fade into a non-propagating electromagnetic field, which we call the evanescent wave. The penetration depth d of this wave is fairly limited, at about two colloid radii, and is given by

$$d = \frac{\lambda_0}{4\pi\sqrt{n_1^2 \sin^2\theta - n_2^2}} , \tag{3.12}$$

where λ_0 is the vacuum wavelength of the laser. This field then excites fluorescent molecules at the surface of the colloids, and we retrieve the emitted light. We can infer the distance between a colloid and the interface z from the intensity a colloid emits, as

$$I(z) = I_0 \exp\left(\frac{-z}{d}\right) , \tag{3.13}$$

where I_0 is the maximal intensity at the interface.

Snapshots of the solution under advection are taken with a camera. Trajectories are then linked from individual successive frames. From the trajectories, displacement distributions are computed, and the transverse diffusion coefficient is then defined as

$$D_y = \frac{\sigma_y^2}{2\tau} , \tag{3.14}$$

where σ_y^2 is the variance of the distribution of displacements in the y-direction (along the channel's wall, perpendicular to the flow), and τ is the time step between two consecutive frames, and is taken as $\tau = 1\,\mathrm{ms}$.

Then, solutions of various concentrations of silica colloids in salted water are tested. It is more intelligible to speak in terms of volume fraction ϕ, that is to say, total volume occupied by colloids, rather than concentration from there on. We define the volume fraction as

$$\phi = \frac{N4/3\pi a^3}{WLH} , \tag{3.15}$$

where N is the total number of colloids in solution. The transverse diffusion coefficients D_y computed from Eq. 3.14 are then recovered for various values of $\phi \in [5 \times 10^{-6}; 4 \times$

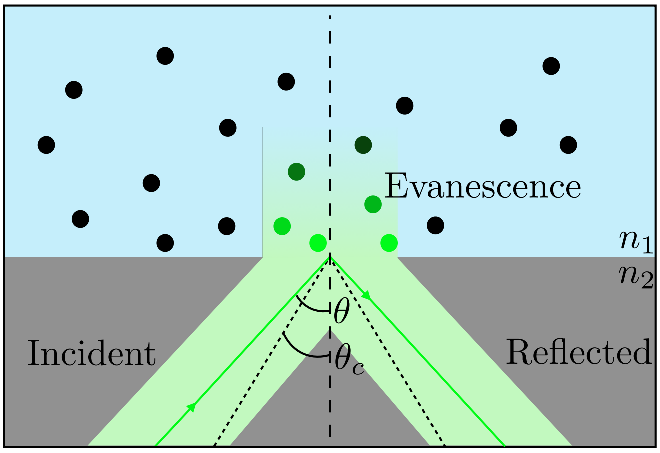

Figure 3.5: TIRFM experimental principle. An incident beam hits the interface with an angle $\theta > \theta_\mathrm{c}$. It is almost entirely reflected away from the interface. A small portion of the beam, however, does penetrate the sample in the form of an evanescent wave. The colloids in the sample can be visualised owing to their fluorescence when they are illuminated by the evanescent wave and their altitude is then recovered from the light intensity they emit.

10^{-5}] and $\dot{\gamma}$ and shown in Fig. 3.6.

Because we work with extremely dilute solutions (below the limit of $\phi = 0.05$ above which the viscosity of the fluid becomes dependant on particle concentration [58]), and as hydrodynamic interactions decay in $\propto 1/r$ inter-colloid, and $\propto 1/r^3$ between a colloid and a wall, we should not expect any visible change on the dynamics of the system within that range of volume fraction. However, experimental results seem to point towards a shear rate and volume fraction dependence on D_y (Fig. 3.6, bottom panels). Indeed, for high enough volume fractions ($\phi = 3.7 \times 10^{-5}$, or 0.000037% of total volume occupied by colloids), D_y becomes larger than the bulk diffusion coefficient D_0 at $D_0 = 3.9\,\mu\mathrm{m}^2.\mathrm{s}^{-1}$, and this effect is exacerbated by higher shear rates.

As puzzling as this result is, there are a few potential causes that can easily be ruled out by carrying out numerical simulations. This has already been done for the case of electrostatic interactions by Élodie Millan as one of her thesis projects.

Numerical methods

As our collaborators in ESPCI hypothesised a hydrodynamic effect causing the enhancement in transverse diffusion D_y, I conduct all the forthcoming simulations using the RigidBlobWalls [20] again. This package, as a reminder, is optimised for the simulation

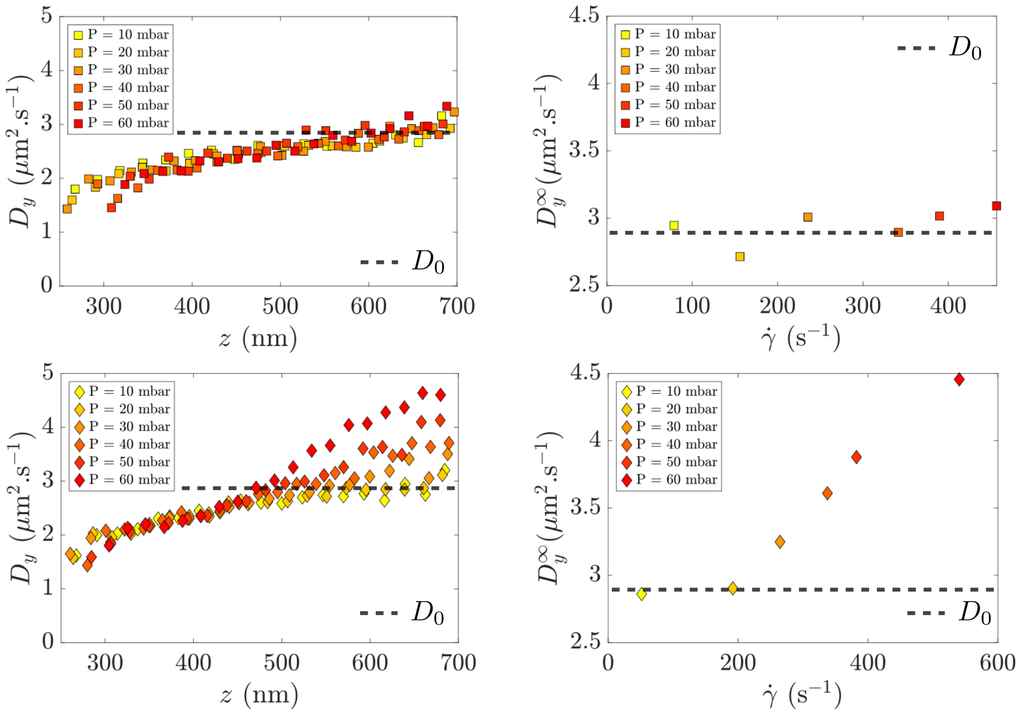

Figure 3.6: Transverse diffusion coefficients computed for $\phi = 5.4 \times 10^{-6}$ (top) and $\phi = 3.7 \times 10^{-5}$ (bottom) and for various pressure gradients ΔP or equivalent shear rates $\dot{\gamma}$ (in colours). Left panels show the diffusion at binned heights z above the channel wall. We notice a strong dependence of D_y on z in both cases, with a drop in diffusion as the colloid diffuses closer to the wall, which is to be expected from Brenner and Faxen's laws. However, once the colloid moves away from the wall, D_y picks up and may even reach higher values than the bulk value $D_0 = 3.9\mu\text{m}^2.\text{s}^{-1}$ (in horizontal dashed lines). This effect seems even more noticeable with higher $\dot{\gamma}$, as shown on the bottom right panel, where D_y^∞ indicates the diffusion coefficient very far from wall interactions. Adapted from [55].

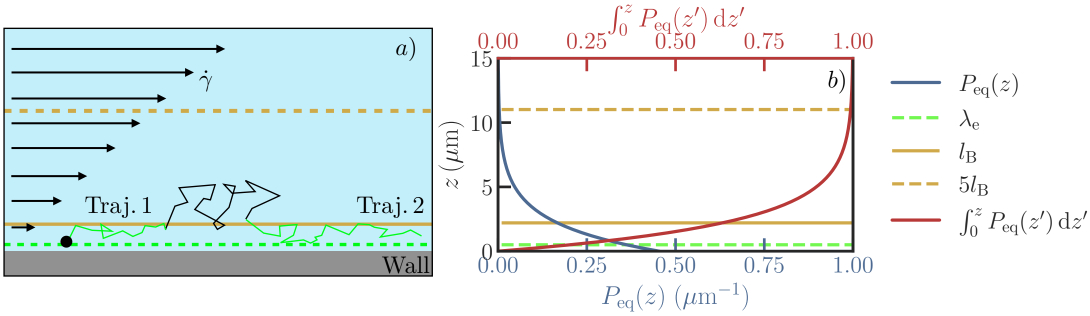

Figure 3.7: Simplified numerical simulation setup. **a)** Colloids are initialised in position in a box of dimensions $L_x \times L_y \times 5l_{\mathrm{B}}$, with l_{B} the Boltzmann constant. The box is periodic in the x and y directions, and has a confining plane at $z = 0$. A shear flow of shear rate $\dot{\gamma}$ is applied in the x direction. Trajectories are recorded for a total time of $T = n_{\mathrm{steps}} \times \Delta t$. Trajectories crossing the evanescent limit λ_{e} are killed, and trajectories re-entering below λ_{e} are tracked as new trajectories if they exist for a long enough time. Data treatment is then done to extract diffusion coefficients from these trajectories. **b)** Equilibrium distribution of a perfect gas of colloids in solution, as predicted by the Gibbs-Boltzmann distribution (blue, Eq. 3.16) and corresponding cumulative sum (red). The evanescent limit λ_{e} is represented in green. This limit is the maximum height at which experimentalists can retrieve data from the colloids. The Boltzmann length l_{B} is represented in a full yellow line. We estimate that 99% of the colloids lie under $5l_{\mathrm{B}}$ (yellow dash line), which is why we choose this height as the upper bound to ensure minimal evaporation of the colloids, and thus a well-defined volume fraction.

of colloids in hydrodynamic and electrostatic interactions, and allows both wall effects and shear flows. The task involves reproducing the TIRFM setup numerically, generating trajectories in the very same conditions as the experiments, and extract similar quantities from these trajectories. The point is to obtain data from a perfectly controlled environment, with perfect tracking of every colloid's position, and with all interactions known.

Simply put, a simulation has four main steps:

Parametrisation. The user provides the physical parameters (buoyant mass of a colloid Δm, colloid radius a, shear rate $\dot{\gamma}$, fluid viscosity η, thermal energy $k_{\mathrm{B}}\Theta$, Debye length l_{D}, amplitude of the electrostatic repulsion potential A, evanescent limit λ_{e}) and simulation parameters (time step Δt, total number of steps n_{steps}, periodic lengths L_x and L_y). This is done using the `inputfile.dat` and `parameters.py` scripts. The `parameters.py` script then reads `inputfile.dat` to convert, calculate, and pass parameters to the other scripts.

As we discussed in the previous paragraphs, the two key quantities we must have full control over are the shear rate $\dot{\gamma}$ and the volume fraction ϕ. The former is simply given as an input to the code in `inputfile.dat`, whereas the latter must be input indirectly in the form of total number N of *blobs*, which in this case are equivalent to colloids.

Initialisation. `sample_initial_positions_gb.py` reads from `parameters.py` the array of ϕ values to be probed. The volume fraction is actually ill-defined in our system if considered without care. Indeed, the simulation code is built to either work in triply periodic boundaries, or doubly periodic in (x, y) and half-infinite in z with a confining plane at $z = 0$. This means the maximum height above which colloids do not escape is set physically, not through user input directly. We can only decide the initial positions of the colloids, but without proper care once the system reaches equilibrium, colloids diffuse in the vertical direction to preferential positions. We recall from Chapter 3 that the distribution of the positions in a perfect gas solution is predicted from the Gibbs-Boltzmann equation

$$P(z) = \frac{1}{Z} \exp\left(\frac{U(z)}{k_\mathrm{B}\Theta}\right) , \tag{3.16}$$

with Z a normalisation factor, and $U(z)$ the total potential energy, as

$$U(z) = A \exp(-\frac{z}{l_\mathrm{D}}) + \frac{z}{l_\mathrm{B}} , \tag{3.17}$$

with A the dimensionless magnitude of the electrostatic repulsion potential, l_D the Debye length and l_B the Boltzmann constant. All these quantities are defined in Chapter 3. The distribution of colloids can be integrated to show that about 99% of the colloids lie under $5l_\mathrm{B}$, and about 60% below l_B (Fig. 3.7b)). Thus, the problem is twofold: if we choose to lower the simulation cost by using a small box of height H_box with $H_\mathrm{box} < 5l_\mathrm{B}$ to place our colloids at the beginning of the simulations, the solution will inevitably homogenize to a lower volume fraction after some simulation time (Fig. 3.8). Furthermore, one cannot simply choose a volume fraction for the whole box and homogeneously place the colloids in that box. Colloids diffuse in the vertical direction and reach the Gibbs-Boltzmann distribution, which is not homogeneous (Fig. 3.7 b)). Thus, the volume fraction is higher close to the wall and lower far from it (Fig. 3.7 b)). To compute the true volume fraction ϕ_{5l_B} we must feed the code for the whole $5l_\mathrm{B}$ slab in order to retrieve the target ϕ within the λ_e slab, we use

$$\phi_{5l_\mathrm{B}} = \frac{\phi\lambda_\mathrm{e}}{5l_\mathrm{B}\frac{1}{Z}\int_0^{\lambda_\mathrm{e}} P_\mathrm{eq}(z)\mathrm{d}z} . \tag{3.18}$$

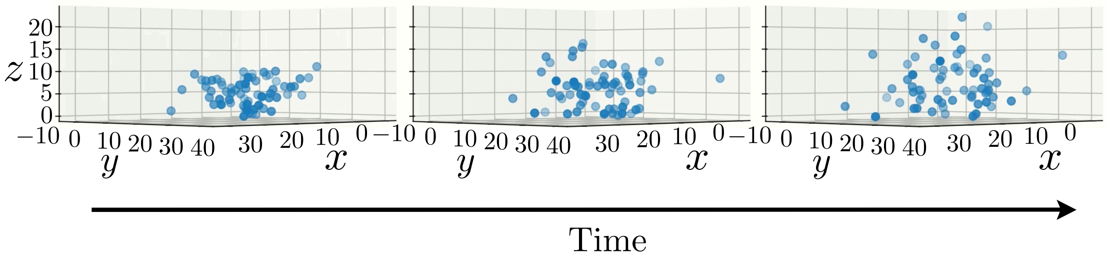

Figure 3.8: Numerical simulation results at initialisation and through time, for a bad initialisation of the system. Colloids were laced in a 10 µm tall box and left to thermalise with $l_\mathrm{B} = 1.1 \times 10^{-2}$ m. As expected, colloids escape the boundaries of the original box from thermal fluctuations, thus effectively reducing the volume fraction inside the box.

3

Then, we may simulate the whole $5l_\mathrm{B}$ slab and fill it with the adequate amount of colloids to reach the desired ϕ. Given that, for our system, $5l_\mathrm{B} = 5.5 \times 10^{-2}$ m, and the TIRFM observation window is of 22 µm × 22 µm, we estimate the total amount of blobs in a simulation at $N = 3.7 \times 10^{7}$ to reach a reasonable target volume fraction $\phi = 4 \times 10^{-4}$. This is clearly much too high, and we must find a workaround.

A very simple way of doing so is to artificially lower l_B. By taking a much higher value for the gravitational acceleration, at $g = 5000 \times g_\mathrm{Earth}$ where $g_\mathrm{Earth} = 9.81\,\mathrm{m.s^{-2}}$, the new Boltzmann constant is $l_\mathrm{B} = 2.20 \times 10^{-6}$ m, which in turn gives a simulation of $N = 152$ colloids. This is much more reasonable.

`sample_initial_positions_gb.py` creates, in the current working directory, a new directory for each value of ϕ. In each new directory, it writes all the simulation scripts. Finally, and most importantly, it computes the initial colloid positions within the simulation box from the Gibbs-Boltzmann distribution of Eq. 3.16 using a simple exclusion algorithm, and writes them to `sphere_array.clones`. This step is crucial to ensure that the system is prethermalised, and avoids a potentially costly prethermalisation step. This script can also be used to verify that the Gibbs-Boltzmann distribution is respected and that the system is properly thermalised.

Simulation. The script `multi_bodies.py` calls the entire simulation pipeline and launches the shear flow on the pre-thermalised colloidal suspension. It writes the trajectories at each time step to `run_blobs.sphere_array.config`. Other relevant simulation data, such as failed iterations and total computation time, are written to auxiliary output files.

Data treatment. From the trajectories stored in `run_blobs.sphere_array.config`, the trajectories are parsed, and the particle displacements are computed.

We take care to ensure we limit our studies to the same field of view as the experiments use. This means that colloids may escape the evanescent wave limit λ_{e} beyond which experimentalists cannot retrieve data do exist in simulation, but that we kill in the treatment process their trajectories once they escape λ_{e}. Furthermore, if they re-enter the field of view below λ_{e} for long enough time durations, their trajectories are treated as new trajectories by the treatment code. MSDs are computed from displacements on each trajectory and then binned by average height.

Diffusion quantities are then extracted from the MSD if, and only if, a diffusive regime is identified, namely beyond the transverse diffusion time τ_{D} and when the log-log fits yield a time exponent equal to 1.

All parametrisation, initialisation, and treatment codes are available on appendix. Simulation codes are available on the GitHub repository of the authors `https://github.com/stochastichydrotools/rigidmultiblobswall`.

Results

Finally, we can compare the results of the simulations to the experimental results. We compute the renormalised transverse diffusion coefficient D_y/D_0 as a function of average trajectory height for various values of ϕ and $\dot{\gamma}$ (Fig. 3.9). We notice that, in contrast to the experimental results, no clear dependence of D_y on ϕ or $\dot{\gamma}$ is visible. The overall tendency of D_y/D_0 shows a slight increase close to the contact with the wall at $z + a =$ 55nm, and quickly reaches a saturation around 1. This is in agreement with the expected behaviour of a Brownian solute in a viscous flow, and with Brenner and Faxen's laws. The absence of any clear dependence on ϕ and $\dot{\gamma}$ suggests that the experimental results may be due to an artefact, and that the observed enhancement of D_y may not be due to hydrodynamic interactions. This is in line with the fact that, at such low volume fractions, hydrodynamic interactions should be negligible. However, it is worth noting that experiments and numerical data were treated differently. Indeed, in experiments, trajectories were first linked from different frames, and displacements were computed from these trajectories. In simulations, we directly access the whole labeled trajectories, thus ruling out any potential tracking artefact.

Furthermore, the simulations were run with a much higher value of g than in experiments, which may have an influence on the results. However, we do not expect this to be the case, as the Boltzmann distribution of colloids in the vertical direction is well respected in simulations, and thus we are confident that the system is properly thermalised and interactions should not be strongly modified.

Finally, it is worth noting that the diffusion coefficient D_y itself has different definitions between our simulation results and the experiments. In experiments, D_y is computed from the variance of the distribution of displacements in the y-direction at a fixed time step τ, as shown in Eq. 3.14. This means that there is no proper verification of a diffusive regime before extracting a diffusion coefficient. If the system is not in a diffusive regime, the extracted coefficient may not be meaningful. In simulations, we compute D_y from the slope of the MSD in the diffusive regime, which is a more standard definition of diffusion coefficient.

Ongoing work is still being conducted to understand the origin of this discrepancy. We are currently analysing numerically generated TIRFM-like videos, which can be found on my GitHub repository ADD THE GIT, with the same tracking and data analysis as the experiments used. If the effect then becomes visible, it would suggest that the effect is due to a data treatment artefact. However, if the effect is not visible, more investigations would be needed. We could for instance verify the actual volume fractions in experiments to fully rule out hydrodynamic effects.

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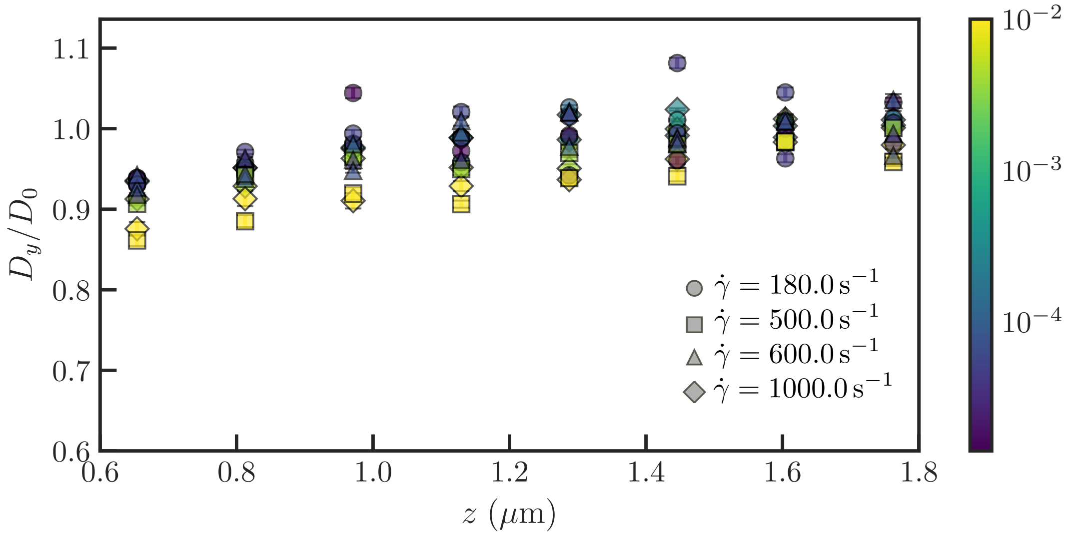

Figure 3.9: Renormalised transverse diffusion coefficient D_y/D_0 as a function of average trajectory height. Simulations were run once for each $(\phi, \dot{\gamma})$ couple and for $N_{\mathrm{steps}} = 50000$, with time step $\Delta t = 0.0001\,\mathrm{s}$, fluid viscosity $\eta = 0.001\,\mathrm{Pa.s}$, colloid radius $a = 55\,\mathrm{nm}$, buoyant mass $\Delta m = 1.92 \times 10^{-16}\,\mathrm{kg}$, gravity acceleration $g = 5000 \times 9.81\,\mathrm{N.m}^{-2}$, thermal energy $k_{\mathrm{B}}\Theta = 4.14 \times 10^{-21}\,\mathrm{J}$, and cutoff evanescent limit $\lambda_{\mathrm{e}} = 1800\,\mathrm{nm}$. The colormap represents volume fractions. Disks correspond to shear rate $\dot{\gamma} = 100.0\,\mathrm{s}^{-1}$, squares to $\dot{\gamma} = 500.0\,\mathrm{s}^{-1}$, triangles to $\dot{\gamma} = 600.0\,\mathrm{s}^{-1}$ and diamonds to $\dot{\gamma} = 1000.0\,\mathrm{s}^{-1}$. The overall tendency of D_y/D_0 shows a slight increase close to the contact with the wall at $z + a = 55$nm, and quickly reaches a saturation around 1. No clear effects of the shear rate are visible.

Chapter 4

Activity in Brownian Motion

4.1 Introduction

The final part of this manuscript is dedicated to the study of active matter and its interplay with Brownian motion. This subject was actually the motivation for my PhD thesis, and the first part of this Chapter is dedicated to my first introduction to the field of active matter. I begin with a full introduction to what Active matter is, and how it is typically modeled in the most popular algorithms.

The second part, which came much later in my PhD, treats the problem of active Taylor-Aris dispersion. The notion of Taylor-Aris dispersion is described in the previous Chapter. For the sake of clarity, I thus invite the reader to consult its developments before reading this part. We conducted this work in tandem with Marc Lagoin, Guirec de Tournemire, Ahmad Badr and Antoine Allard, and published it in PNAS.

4.1.1 Activity

Authors define active matter as one or both of the following:

1. "energy-consuming particles" : individual units that extract free energy from internal or external sources and convert it to motion, forces, or mechanical stresses (this is the definition I use in this manuscript) . This work is not necessarily expressed in the form of self-propelling [59][60][61] [62]. They are thus inherently out-of-equilibrium systems;
2. "interacting particles" : individual self-propelled units which interact locally and exhibit emergent, collective behaviour, such as hydrodynamic interactions (swimming microorganisms, squirmers, active droplets...)[61], phoretic interactions (cat-

alytic Janus colloids, some chemotactic bacteria...)[63], mechanical force transmission (cellular tissue, cytoskeleton, active gels...)[59], and even social interactions (human crowds, bird flocks, fish schools, autonomous drones...)[64], to only cite a few. Some of these interactions can be accurately represented with minimal models, which approximate them as alignment interactions with some noise, such as the Vicsek model [65]. Certain authors distinguish these instances of interacting active matter from the first I mentioned [62].

Numerous examples of active systems exist, and these systems can be found at any scale (Fig. 4.1). I find it convenient to not only classify them by size but also by typical Reynolds number, as most of them will typically use their surrounding energy source to move in space.Interestingly, the definition of active matter is loose enough to even include human beings, robots and even large aircrafts. This thesis articulates around notions of small systems, and I thus refer to these systems form there on.

We may for example cite the very famous example of *Chlamydomonas Reinhardtii*. These tiny algae will be the basis of our publication at the end of this Chapter. They measure only a few tens of microns at most. However, through the beating of the flagella at their front, they are able to self-propel in a fluid at the surprisingly high speed of ten body lengths per second. Yet, this is not sufficient for *Chlamydomonas* to be considered high-Reynolds systems ($R_\mathrm{e} \gg 1$), as they typically reach a Reynolds number of about $R_\mathrm{e} = 1 \times 10^{-4}$. They thus still evolve in a very viscous medium. Furthermore, these algae are able to orient towards or away from light sources through the use of an eye spot, which is able to sense both a light's wavelength and its intensity. Under certain conditions, their behaviour may completely change from a swimmer with random and erratic reorientations, to phototactic, to a glider, stuck to the boundary of its observation box, and only slowly moving in its sticking plane.

4.2 Classes of microswimmers

4.2.1 Biological swimmers

4.2.2 Artificial swimmers

4.3 Modelling approaches

Multiple techniques exist to simulate active matter. I describe here the ones I have come to know or use during my thesis, starting with non-interacting simulations (ABPs, RNTs) [62], and then more complete and complex methods involving interactions and finite-size effects (Squirmers, Mobility-based methods and fully resolved hydrodynamic

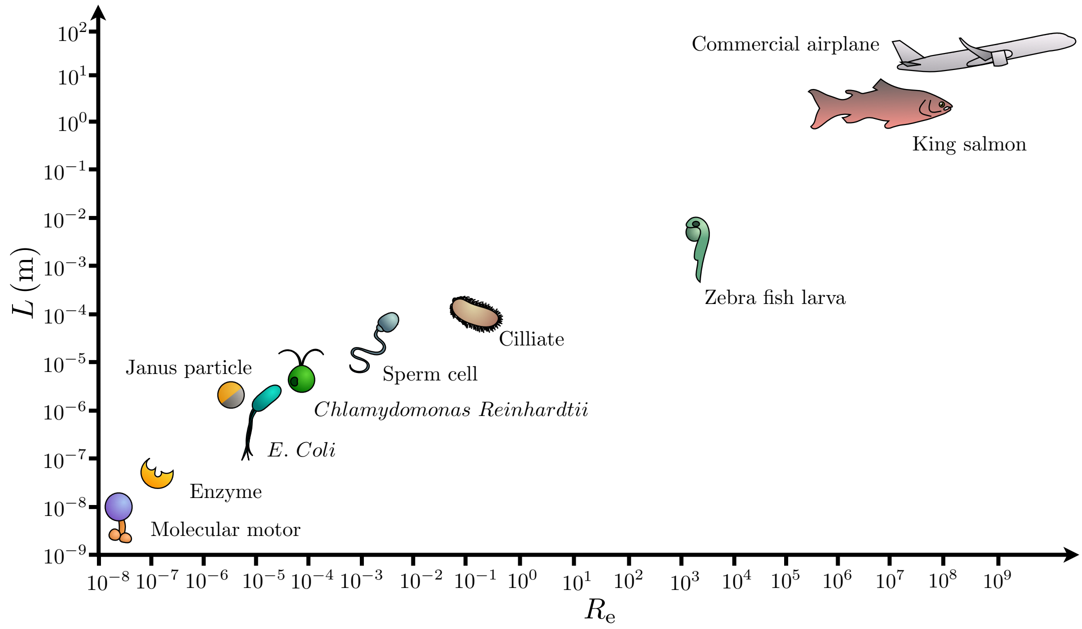

Figure 4.1: Comparison of typical length scales and Reynolds numbers for various active systems.

methods). I also provide in appendix and on the GitHub repository ADD GITHUB a simple Jupyter file to simulate ABPs, RNTs and to compare them to the passive case. This file also contains a very small exploration of chiral Brownian particles.

4.3.1 Active Brownian Particles

For comparison purposes, let us recall that a purely passive colloid undergoes thermal diffusion. This thermal diffusion affects both the translational degrees of freedom (x and y for a two-dimensional case) and the rotational degree of freedoms (ϕ for a two-dimensional case). We define from the Stokes-Einstein relation the translational thermal diffusion coefficient D_{T} as:

$$D_{\mathrm{T}} = \frac{k_{\mathrm{B}}\Theta}{6\pi\eta a} \ , \tag{4.1}$$

where $k_{\mathrm{B}}\Theta$ is the thermal energy of the bath, η the fluid visicosity and a the colloid radius. The rotational diffusion coefficient D_{R}, however, is proportional to the colloid's volume:

$$D_{\mathrm{R}} = \frac{k_{\mathrm{B}}\Theta}{8\pi\eta a^3} = \tau_{\mathrm{R}}^{-1} \ , \tag{4.2}$$

with τ_{R} a characteristic reorientation timescale.

We have already stated that we can describe the motion of the passive colloid with the following Langevin equations for the overdamped limit:

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= \sqrt{2D_{\mathrm{T}}}\xi_x(t) \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= \sqrt{2D_{\mathrm{T}}}\xi_y(t) \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} &= \sqrt{2D_{\mathrm{R}}}\xi_\phi(t) \ . \end{aligned} \tag{4.3}$$

where $\xi_i(t)$ white Gaussian noises of correlation $\delta(t)$. This process is heavily reliant on the thermal diffusion coefficients D_{T} and D_{R}, which are themselves proportional to temperature and inversely proportional to colloid radius and fluid viscosity. This means that thermal Brownian motion is strongly hindered in the case of larger colloids, where larger means of the order of a few tens of microns. For example, if we take for comparison with later cases a colloid of a size that is comparable to a *Chlamydomonas*, that is to say $a = 10\mu\mathrm{m}$, we get $D_{\mathrm{T}} = 2.2 \times 10^{-3}\mu\mathrm{m}^2.\mathrm{s}^{-1}$. We interpret that by pure thermal diffusion, a colloid of this size would need tens of hours to explore an area equivalent to its own size.

Lucky, as we discussed already, swimmers circumvent this limitation by collecting energy from their surroundings to self propel amongst other strategies. A very simple model used to represent a swimmer with an ability to self-propel with a given velocity $\vec{v}$ is the Active Brownian Particle (APB) model. We describe this model with the following Langevin equations:

$$\begin{aligned}
\frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D_\mathrm{T}}\xi_x(t) \\
\frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D_\mathrm{T}}\xi_y(t) \\
\frac{\mathrm{d}\phi}{\mathrm{d}t} &= \sqrt{2D_\mathrm{R}}\xi_\phi(t)\ .
\end{aligned} \tag{4.4}$$

We notice that the swimmer is still undergoing thermal diffusion for the rotational degree of freedom, with a characteristic time τ_R after which the swimmer "forgets" in which direction it was heading. This memory effect can be noticed at very short times

Case	D_T	Persistence time	D_eff
Passive BM ($a = 0.5\,\mu$m)	0.44	–	0.44
ABP ($a = 0.5\,\mu$m)	0.44	0.77 s	0.82
Passive BM ($a = 10\,\mu$m)	2.2×10^{-2}	–	2.2×10^{-2}
ABP ($a = 10\,\mu$m)	2.2×10^{-2}	6250 s	3.0×10^{3}

Chiral ABPs

Some microswimmers or clusters of microswimmers have irregular shapes or asymmetrical propulsion mechanisms.

While the ABP translation mechanisms remain unchanged, one can intriduce chiral behaviour in the ABP model by adding a phase in the rotational degree of freedom, as follows:

$$\begin{aligned}
\frac{\mathrm{d}x}{\mathrm{d}t} &= v\cos[\phi(t)] + \sqrt{2D_\mathrm{T}}\xi_x(t) \\
\frac{\mathrm{d}y}{\mathrm{d}t} &= v\sin[\phi(t)] + \sqrt{2D_\mathrm{T}}\xi_y(t) \\
\frac{\mathrm{d}\phi}{\mathrm{d}t} &= \omega\sqrt{2D_\mathrm{R}}\xi_\phi(t)\ .
\end{aligned} \tag{4.5}$$

It is therefore possible to play with the phase to obtain different behaviours, from shrinking spirals to circles to expanding spirals for the two-dimensional case.

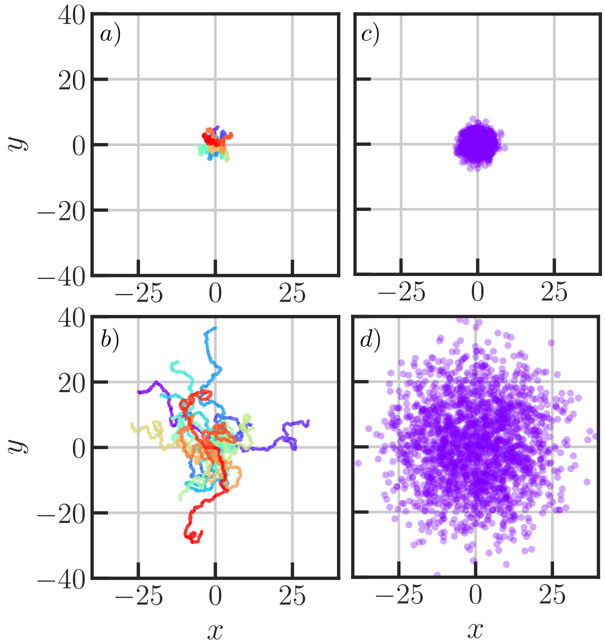

Figure 4.2: Comparison of passive and active Brownian colloid trajectories. Panels **a)** and **c)**: Twenty stacked trajectories obtained from discretising Eqs. 4.3 and 4.5 respectively, with $\Delta t = 0.05$, $n_{\mathrm{steps}} = 5000$, $D_{\mathrm{T}} = 0.05$, $D_{\mathrm{R}} = 0.3$, $v = 0.1$, with all units in simulation units. Colours are only used for readability and have no physical meaning. Panels **c)** and **d)**: final position of 2000 simulations for the passive and active Brownian particle models, respectively.

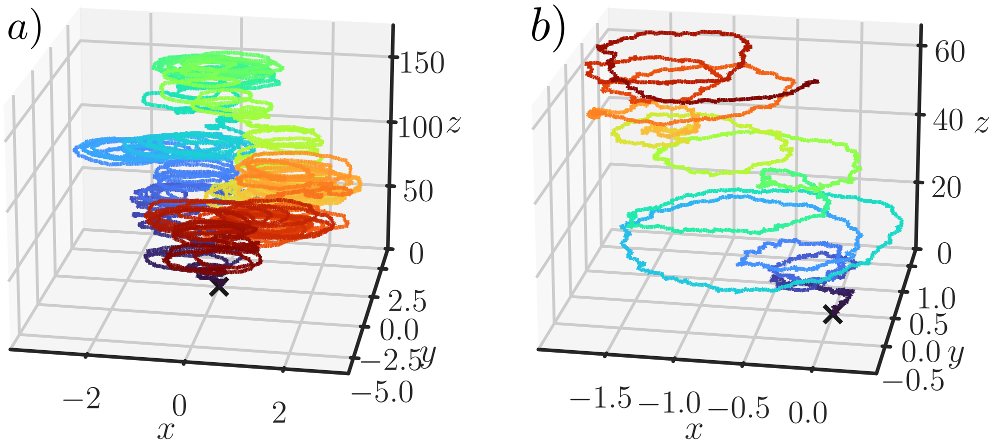

Figure 4.3: Simulated trajectories from the discretised Eqs. 4.7. **a)** Full trajectory for $n_{\text{steps}} = 80000$, $\Delta T = 0.01$, $D_{\text{T}} = 0.005$, $D_{\text{R}} = 0.01$, $v = 1.0$, $\boldsymbol{\Omega} = 1.0$ and $\omega = \frac{\pi}{12}$. All units are in simulation units.

Helical motion can be described using a three-dimensional ABP model, such as

$$\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = v\mathbf{n} + \sqrt{2D_t}\,\boldsymbol{\xi}_{\mathbf{T}}(t), \tag{4.6}$$

$$\frac{\mathrm{d}\mathbf{n}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \mathbf{n} + \sqrt{2D_r}\,\boldsymbol{\xi}_{\mathbf{R}}(t) \times \mathbf{n}, \tag{4.7}$$

with $\mathbf{r}$ the position vector $\mathbf{r} = (x, y, z)$, and $\mathbf{n}$ the orientation vector, which must be normalised, as $\mathbf{n} = (n_x, n_y, n_z)$, $|\mathbf{n}| = 1$, and $\boldsymbol{\Omega}$ is the angular velocity vector defined as

$$\boldsymbol{\Omega} = \omega\,\mathbf{e}_z = \begin{pmatrix} 0 \\ 0 \\ \omega \end{pmatrix} , \tag{4.8}$$

for a rotation about the z axis (Fig. 4.3).

4.3.2 Run-and-tumble

4.3.3 Squirmers

Pullers-Pushers

4.3.4 Mobility based methods

4.3.5 Fully resolved hydrodynamic methods

4.4 Active Brownian Particles in fluid flows

4.4.1 Context

4.4.2 Experimental setup

4.4.3 Numerical model

4.4.4 Results

4.5 Conclusion and perspectives

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